Texture analysis on computed tomography (CT) data from renal cell carcinoma cases

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LIST OF PUBLICATIONS

Yuan Liang, Abraham G. Campbell, and Sourav Bhattacharjee, *"Renal cell carcinoma texture analysis with the KiTS23 dataset."* (under review) in the Advances in Urology, 2025.

Fangyijie Wang, **Yuan Liang**, Sourav Bhattacharjee, Abey Campbell, Kathleen Curran, and Guenole Silvestre, *"Fusing Radiomic Features with Deep Representations for Gestational Age Estimation in Fetal Ultrasound Images."* (Accepted) MICCAI: The 28th International Conference on Medical Image Computing and Computer Assisted Intervention, 2025.

Yuan Liang, Zixiang Xu, Sanaz Rasti, Soumyabrata Dev., and Abraham G. Campbell, *"On the Use of a Semantic Segmentation Micro-Service in AR Devices for UI Placement."* In The IEEE CTSoc International Conference on Games, Entertainment & Media, 2022.

Xu, Zixiang, **Yuan Liang**, Abraham G. Campbell, and Soumyabrata Dev., *"An Explore of Virtual Reality for Awareness of the Climate Change Crisis: A Simulation of Sea Level Rise."* In 2022 8th International Conference of the Immersive Learning Research Network (iLRN), pp. 1-5. IEEE, 2022.

Zheng, Mengya, Rosemary J. Thomas, Xingyu Pan, Zixiang Xu, and **Yuan Liang**, *"Augmenting Feature Importance Analysis: How Color and Size Can Affect Context-Aware AR Explanation Visualizations?"* In 2022 IEEE International Symposium on Mixed and Augmented Reality, 2022.

Zixiang Xu, **Yuan Liang**, Abraham G. Campbell, and Soumyabrata Dev, *"Climate Crisis in Virtual Environments: Exploration and Evaluation of Virtual Reality and Mixed Reality for Climate Change Education in Sea Level Rise Simulation."* In 8th International Conference of the Immersive Learning Research Network (iLRN), 2022.

ABSTRACT

This report presents the progress, findings, and planned continuation of a PhD project investigating the integration of deep learning and radiomics for texture analysis of renal tumours using computed tomography. The overarching goal is to establish whether multimodal fusion can enhance both the diagnostic accuracy and interpretability in renal tumour classification.

A comprehensive literature review and systematic analysis were conducted to evaluate existing approaches in CT-based renal tumour classification. The review revealed two major research gaps: (1) the limited use of DL for direct texture analysis of renal tumours, and (2) the lack of systematic investigations into the fusion of radiomic features with DL representations. To address these limitations, a novel feature fusion framework was developed that integrates handcrafted radiomic descriptors with high-level deep features, aiming to balance performance and transparency.

This framework was first validated in a 2D ultrasound setting for gestational age estimation, confirming its technical feasibility and potential for generalisation across modalities. Building on this success, the approach will be extended to 3D CT data for renal tumour subtype and malignancy classification. Public datasets such as KiTS23 and TCGA–KIRC will serve as the primary data sources to ensure reproducibility and external validity.

Future work will systematically compare different multimodal fusion strategies, including numerical, attention-based, and vision–language integrations, and incorporate metadata through both tabular encoders and natural-language representations. Benchmarking will be performed against radiomics-only and CT-only baselines. Interpretability studies will move beyond post hoc explanation tools, focusing instead on visualising the role of radiomic and metadata features in the decision process to provide clinically grounded transparency. The final phase will consolidate these contributions into a comprehensive doctoral thesis and a series of publications.

CHAPTER

ONE

INTRODUCTION

1.1 Research Context

Kidney cancer accounted for approximately 431,288 new cases worldwide in 2020, ranking ninth in incidence among men and fourteenth among women [1, 2]. Renal cell carcinoma (RCC) represents more than 90% of these cases [3, 4, 5, 6]. Increased utilisation of sensitive imaging modalities such as CT and MRI has contributed to a higher detection rate and earlier diagnosis [7, 8].

Traditional oncologic imaging relies heavily on subjective interpretation, whereas radiomics offers quantitative, reproducible descriptors that capture tumour heterogeneity and morphology [9, 10]. By transforming segmented regions into high-dimensional texture features [11, 12], radiomics supports machine learning (ML) and deep learning (DL) models for tasks including subtype classification, gene expression prediction, and prognostication [11, 12, 13, 14]. DL can further perform texture analysis directly from images [15], although its performance depends on segmentation quality [11, 12]. As a result, radiomics has become an influential tool in non-invasive tumour characterisation [13].

ML and DL techniques continue to transform medical imaging by learning complex patterns with increasing accuracy and reduced manual intervention [15, 16]. ML identifies discriminative relationships within data [17, 13], while DL leverages hierarchical feature representations [18], often achieving superior performance [18, 16, 15]. In RCC imaging, these methods have been applied to classification, segmentation, grading, and outcome prediction [16, 15, 11, 12, 13, 14]. However, DL-based texture analysis of renal tumours remains limited, with most studies relying on handcrafted radiomics or using DL primarily for segmentation [19]. Direct comparisons between radiomics-based ML pipelines and DL models are scarce, and multimodal frameworks combining handcrafted and learned representations remain underexplored.

Radiomics features provide interpretable descriptors of tumour appearance [11, 12], whereas DL features are powerful but less transparent [18, 16, 15]. Their complementary nature suggests that combining both modalities may yield synergistic benefits [20, 21, 22]. Existing fusion strategies—including early concatenation, attention mechanisms, and multi-branch networks—indicate potential improvements in accuracy and interpretability [20]. Yet, systematic evaluation of these methods in 3D

CT-based renal tumour analysis is limited.

This project investigates multiple fusion approaches for 3D CT classification, assessing accuracy, robustness, and interpretability. Beyond numeric fusion, an emerging direction is to leverage vision–language models (VLMs) by converting radiomics features into natural-language descriptions. VLMs integrate image and text information [23, 24], and textual prompting has been shown to enhance model performance in related imaging tasks [25]. In this work, radiomics values and relevant clinical information are translated into descriptive textual prompts and supplied to a pretrained VLM alongside CT volumes, enabling richer cross-modal representation learning.

Recent evidence demonstrates that incorporating textual descriptions of imaging features improves performance and generalisability, as shown by Ra *et al.* for breast tumour classification using an LLM-enhanced radiomics pipeline [26]. Building on this concept, the proposed framework unifies 3D imaging and radiomic information within a VLM for renal tumour classification. This study evaluates the advantages of such textual fusion relative to conventional numeric fusion, focusing on predictive accuracy, robustness, and interpretability, with clinician assessment of textual outputs.

The framework further extends to incorporate clinical metadata, which can also be expressed as structured text. Integrating imaging, radiomics, and metadata within a multimodal VLM is expected to enhance diagnostic performance while offering improved transparency in decision-making.

1.2 Research Objectives

This thesis aims to advance the integration of DL and radiomics for texture analysis in medical imaging, with a focus on 3D CT-based renal tumour classification. Renal tumours were selected not only due to the limited adoption of DL in this domain but also because current classification algorithms exhibit a notable performance gap.

Although renal tumours represent a smaller share of global cancer incidence than lung or breast malignancies, they remain clinically important owing to their often asymptomatic presentation and the lack of reliable imaging biomarkers. They also pose inherent imaging challenges, including subtle inter-class differences, anatomical variability, and marked label imbalance across subtypes [27]. The biological heterogeneity of RCC further provides an opportunity to design models that capture fine-grained visual and radiomic variations [28, 29]. However, the scarcity of large, standardised datasets has limited the development of generalisable benchmarks, contributing to the relatively small body of kidney tumour research in medical imaging [30]. At the same time, this gap highlights a timely research opportunity. With increasingly curated CT datasets and progress in multimodal learning, renal tumour analysis stands to benefit from frameworks that combine radiomic, metadata, and imaging information to improve diagnostic accuracy and interpretability.

Accordingly, renal tumour classification offers both a clinically meaningful context and a technically

demanding environment for evaluating advanced fusion approaches. This work addresses key gaps by systematically assessing the performance, robustness, and interpretability of multiple modelling strategies and feature-fusion paradigms.

The specific objectives of this research are as follows:

1. To identify, process, and curate suitable publicly available CT renal tumour datasets (e.g., KiTS[1], TCGA[2]) for model training, validation, and benchmarking, ensuring reproducibility and external generalisation.

2. To extract and interpret handcrafted radiomics features from the curated datasets, and to conduct texture-based classification using conventional ML methods as a baseline reference for subsequent benchmarking.

3. To quantitatively evaluate the performance gain of DL models over traditional radiomics-based ML approaches in texture analysis, identifying strengths and limitations across architectures and feature types.

4. To develop and systematically compare advanced feature-fusion frameworks that integrate radiomics and imaging representations, investigating strategies such as early, late, and attention-based fusion, and designing appropriate loss functions to enhance texture discrimination and model stability.

5. To explore VLM-based fusion as an advanced multimodal learning approach, converting radiomic features into natural language descriptions and aligning them with imaging data through pretrained VLMs. This objective aims to leverage the semantic reasoning capabilities of VLMs for improved interpretability and cross-modal representation learning.

6. To extend the radiomics framework into a multimodal paradigm by incorporating additional information such as patient metadata or clinical descriptors, enabling richer contextual learning and improving generalisability in renal tumour classification.

7. To validate the interpretability and clinical relevance of the proposed models through expert evaluation, assessing whether the model's feature attributions and generated textual explanations align with radiological reasoning and clinical decision-making.

1.3 Research Questions

At the outset of this project, the research was broadly framed around texture analysis of renal tumours, with an initial plan to develop a multimodal segmentation model integrating DL and radiomics features,

[1] https://kits-challenge.org/kits23/
[2] https://www.cancer.gov/ccg/research/genome-sequencing/tcga

followed by 3D visualisation to support clinical interpretation of tumour morphology.

However, insights gained during the literature review prompted a critical refinement of this direction. Radiomics features have historically been used for classification and prediction tasks—such as subtype identification, grading, and prognostic modelling—rather than for segmentation [9, 10, 11, 12]. A clear mismatch emerged between the nature of radiomics and the objectives of segmentation, which, while essential as a pre-processing step for region-of-interest extraction, is not itself a radiomics-driven task [31]. Moreover, DL has already become the dominant approach for medical image segmentation, leaving limited scope for methodological novelty in this area [31, 19]. In contrast, the use of DL for radiomics-style texture analysis, including renal tumour classification and outcome prediction, remains relatively underexplored, revealing a promising research gap.

Consequently, the project focus shifted towards investigating DL-based texture analysis for renal tumour CT imaging, with an emphasis on assessing its feasibility and performance relative to traditional radiomics-based ML approaches. This adjustment better aligns with the strengths of radiomics and targets an area with substantial potential for advancement.

This refinement also highlighted opportunities for multimodal feature fusion, enabling the combination of handcrafted radiomic descriptors with learned DL representations. Such fusion strategies, rather than treating the modalities as competing, allow models to capture both abstract features and clinically interpretable quantitative information [20]. Beyond improving classification accuracy, this integration supports model transparency, which is vital for clinical acceptance and decision support in renal tumour analysis.

Overall, these developments reflect an iterative, evidence-driven redefinition of the research strategy, grounded in methodological alignment and clinical relevance. Guided by the research gaps identified, this thesis aims to address the following research questions:

- **RQ1:** How can radiomics and DL representations be effectively integrated to improve texture-based classification performance across medical imaging modalities?
- **RQ2:** What are the most effective feature-fusion strategies, including conventional numerical fusion and advanced VLM-based approaches, for aligning radiomic features with imaging representations?
- **RQ3:** To what extent can the proposed multimodal frameworks, incorporating radiomics, imaging, and metadata, enhance accuracy, robustness, and interpretability in 3D CT-based renal tumour classification?
- **RQ4:** How well do the interpretive mechanisms and textual outputs of the proposed models align with radiological reasoning, and how can expert evaluation be used to validate their clinical interpretability?

1.4 Report Overview

In the next chapter (Chapter 2), a comprehensive literature review will be conducted. This will include a systematic review of the current state of the art in applying DL and ML to CT-based texture analysis of renal tumours, an introduction to CT-based radiomics using the PyRadiomics toolkit, a targeted review of relevant computer vision techniques in medical imaging, and a focused discussion on multimodal feature fusion strategies for integrating radiomic features, clinical data, and DL features, including VLMs. As part of the systematic review, publicly available datasets relevant to renal tumour analysis will be summarised and evaluated, and the rationale for selecting specific datasets as training data and external validation in this study will be clearly articulated. These reviews will collectively identify key research gaps and provide the foundation for the research questions and contributions addressed in this thesis.

Chapter 3 introduces the specific problem domains corresponding to the research questions and outlines the methodological approaches adopted to address these challenges. This chapter also provides a summary of the research progress achieved thus far. To date, three key milestones have been reached.

First, a comprehensive understanding of radiomics feature extraction has been developed, and conventional ML techniques have been applied to perform texture-based classification on the KiTS23 dataset. The outcomes from this traditional pipeline serve as a benchmark for evaluating the performance of subsequent DL-based approaches. **Second**, initial experiments have been conducted using state-of-the-art feature fusion strategies on lightweight 2D medical imaging datasets. These experiments provided a preliminary assessment of the feasibility and advantages of integrating handcrafted radiomics features into DL architectures, laying the foundation for more complex multimodal investigations. **Third**, the feature-fusion strategies developed and evaluated in Task 2, together with other state-of-the-art fusion techniques, have been adapted and applied to the 3D CT-based KiTS23 dataset. At this stage, DL and fusion-based architectures were implemented and compared against the Task 1 baseline of traditional radiomics combined with ML. The experiments aim to evaluate how transferring, adapting, and extending these fusion strategies influence classification accuracy, robustness, and interpretability in renal tumour analysis. The findings from this stage provide critical insights for refining multimodal fusion pipelines and informing the subsequent development of VLM-based models.

A detailed research plan, including the proposed workflow and timeline, is presented in Chapter 4.

CHAPTER

TWO

LITERATURE REVIEW

2.1 Systematic Review - Deep Learning and Radiomics in Renal Tumour Classification

This systematic review is currently being prepared for submission to a peer-reviewed journal.

Before presenting the findings of the systematic review, it is important to outline its position within the context of this research project. This review was conceived as an initial step to critically assess the current landscape of ML and DL applications in renal tumour texture analysis before answering the research questions. Tumour classification tasks, such as distinguishing between benign and malignant renal tumours, or subtyping RCC, serve as a relevant and clinically meaningful entry point for evaluating the potential of DL-based texture analysis.

The review aimed to determine how DL compares with traditional radiomics-based ML methods in terms of diagnostic performance, methodological consistency, and data requirements. Classification was chosen for its direct clinical relevance and reliance on texture information, making it an appropriate test case for assessing DL feasibility. The findings showed that radiomics combined with ML remains the dominant paradigm, while evidence for DL is present but limited. This confirmed that DL is feasible for texture analysis but highlighted a clear research gap: DL is not yet widely applied to renal tumour classification, and few studies directly compare DL with radiomics-based ML pipelines. The review also exposed substantial methodological heterogeneity across studies, including variations in CT acquisition phases, feature extraction procedures, model architectures, and evaluation strategies. Such inconsistencies emphasise the need for more rigorous and standardised methodologies. Notably, the absence of studies combining radiomics features with DL representations underscores the potential value of multimodal fusion, motivating a shift in research focus towards unified fusion frameworks.

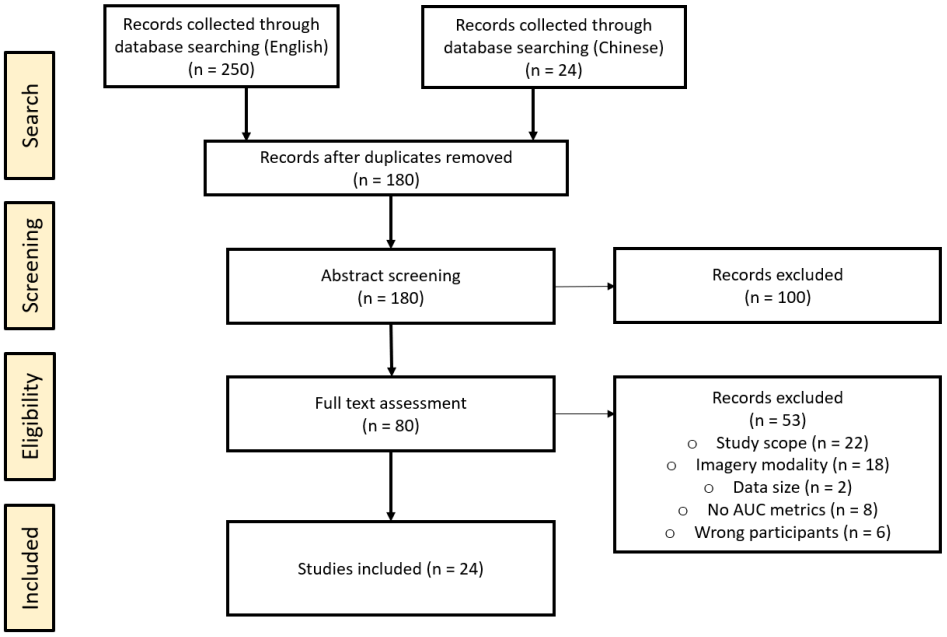

Figure 1: PRISMA Flow-chart of included studies. There are 24 studies included.

2.1.1 Results

A total of 274 records were identified through database searches (250 English-language and 24 Chinese-language articles). Chinese publications were included to ensure comprehensive coverage of rapidly expanding radiomics and DL research from China, with screening supported by a Chinese-speaking author. After removing duplicates, 180 records remained; title and abstract screening excluded 100, leaving 80 for full-text review. Following application of all eligibility criteria, 24 studies were included. These focused specifically on CT-based radiomics or DL methods for renal tumour classification and reported relevant performance metrics. The selection process is summarised in Figure 1.

2.1.2 Private vs Public Datasets

Most studies (79%) relied exclusively on internal institutional datasets. While such datasets benefit from consistent imaging protocols and expert annotation, they limit generalisability and reproducibility. Public datasets are therefore essential for benchmarking and external validation. However, heterogeneity in annotation standards, imaging phases, and data quality persists even within public resources, underscoring the need for larger, well-curated repositories with consistent metadata.

2.1.3 Public Data Source

Nineteen studies used internal datasets, whereas only five incorporated public datasets. The primary public resources were:

- The Cancer Genome Atlas (TCGA),
- The Cancer Imaging Archive (TCIA)[1], and
- The Kidney Tumor Segmentation Challenge (KiTS) datasets.

TCGA provides multi-omics profiles and extensive clinical information across numerous tumour types [32], but no imaging data. TCIA complements TCGA by offering de-identified CT, MRI, and PET imaging [33], including collections that bridge imaging and genomics (e.g., TCGA-KIRC). CT enhancement phase labels, however, are inconsistent and often require manual inspection of DICOM metadata. Limited clinical metadata is available for some collections.

KiTS19, KiTS21, and KiTS23 focus on kidney cancer segmentation [34]. KiTS23 provides increased diversity by including both corticomedullary and nephrogenic phase CT scans. All KiTS datasets supply high-quality 3D segmentations and basic clinical variables (e.g., age, sex, laterality, subtype), though not molecular data. Despite containing around 500 cases, KiTS23 is considered large by medical imaging standards, where expert annotation is costly. Previous work has shown that state-of-the-art models such as nnU-Net, TransUNet, and SwinUNETR achieve strong generalisation using only a few hundred cases [35].

KiTS23 also offers rich multimodal information: voxel-level annotations, radiomics-feasible segmentations, clinical metadata, and extractable deep feature representations. Each CT volume can be decomposed into numerous overlapping 3D patches [36], effectively expanding the training set. Transfer learning and self-supervised pretraining further mitigate dataset size limitations, with studies demonstrating performance gains over training from scratch [37, 38, 36, 39]. Extensive augmentation strategies (rotations, flips, intensity perturbation) provide additional regularisation, and ImageNet-pretrained models have been shown to perform robustly even on datasets of this scale [40]. Collectively, these factors support the suitability of KiTS23 for reliable DL classification research.

Additionally, portions of the TCGA imaging data within TCIA have been annotated as part of the AIAM initiative, where AI-generated segmentations were reviewed by expert radiologists [41]. These standardised and quality-controlled annotations enhance the utility of TCIA for external validation. Consequently, TCIA datasets provide a strong foundation for testing the robustness and generalisability of models trained on KiTS23 across diverse real-world imaging conditions.

2.1.4 Tumour Subtype Classification Performance

Four clinically relevant tasks were identified across the included studies: (1) benign vs. malignant discrimination; (2) RCC subtype classification; (3) AML vs. RCC differentiation; and (4) oncocytoma vs. chromophobe RCC classification. A quantitative performance summary is provided in Table 2.1. While moderate to high accuracy was reported in some settings, performance remains insufficient for

[1] https://www.cancerimagingarchive.net/

routine clinical use, highlighting the potential for DL methods to improve robustness and subtype-specific discrimination.

Table 2.1: Summary of Model Performance by Classification Task

Classification Task	N Studies	Mean AUC	Std Dev	Min AUC	Max AUC
Benign vs Malignant	10	0.809	0.076	0.715	0.960
RCC Subtype Classification	6	0.814	0.068	0.720	0.910
AML (wvf/fp) vs RCC	4	0.871	0.066	0.810	0.955
Oncocytoma vs Chromophobe RCC	2	0.847	0.193	0.710	0.983

2.1.5 Systematic Review Conclusion

Radiomics and ML approaches hold considerable promise for enhancing the non-invasive characterisation of renal tumours using CT imaging. This systematic review highlights a growing body of evidence supporting the feasibility of classifying tumour subtypes and distinguishing benign from malignant lesions with high diagnostic performance. Despite this progress, the field remains methodologically heterogeneous, with variation in classification tasks, feature selection techniques, imaging phases, and evaluation standards. Most studies employed traditional ML algorithms and handcrafted radiomics features, with limited adoption of DL. Corticomedullary and nephrographic phases were most commonly used, reflecting their diagnostic relevance and radiological contrast advantages.

Although several high-quality, large-scale public datasets such as KiTS, TCGA, and TCIA are available, featuring well-annotated, multi-phase CT scans ideal for radiomics research, relatively few studies have utilised them. This underuse limits the reproducibility and generalisability of current findings and slows the pace of clinical translation.

Future research should prioritise the use of larger, multi-institutional public datasets, standardised radiomics pipelines, and increased adoption of DL methods. With greater methodological rigour and data transparency, radiomics has the potential to become an integral component of renal tumour diagnosis and personalised management.

2.2 CT-Based Radiomics Using PyRadiomics

Radiomics enables the extraction of high-throughput quantitative features from medical images, converting them into structured variables for statistical or ML analysis. In renal tumour CT imaging, radiomics provides a non-invasive means of assessing tumour heterogeneity, intensity patterns, and morphology. Among existing radiomics tools, *PyRadiomics* is widely used due to its standardisation, flexibility, and integration with Python [42].

A typical workflow begins with acquisition of contrast-enhanced CT—most often in the corticomedullary or nephrographic phase—followed by pre-processing steps such as voxel resampling,

intensity discretisation, and optional denoising. Segmentation then defines the region of interest, performed manually, semi-automatically, or using DL. PyRadiomics requires spatially aligned image–mask pairs in standard formats (e.g. NIfTI, DICOM).

From the segmented ROI, PyRadiomics extracts first-order statistics, shape descriptors, and texture features derived from matrices such as GLCM, GLRLM, GLSZM, NGTDM, and GLDM. Additional filtered features (e.g. wavelet, LoG) enable multi-scale texture analysis. Because this process can generate several hundred features per case, feature selection is essential. Common methods include variance and correlation filtering, RFE, LASSO, and mutual-information-based selection [43, 44]. Standardisation (e.g. z-scoring) is typically required before modelling.

Selected features are used to train ML classifiers such as SVM, random forests, logistic regression, or gradient boosting, with performance evaluated via accuracy, AUC, sensitivity, and specificity under cross-validation. PyRadiomics adheres to IBSI guidelines, facilitating reproducibility, and integrates seamlessly with Python ML libraries such as `scikit-learn`.

Despite these advantages, radiomics is sensitive to segmentation variability, acquisition differences, and feature instability. These limitations highlight the need for standardised imaging protocols, robust validation, and feature stability assessment.

2.3 Computer Vision in Medical Imaging

In cancer imaging, AI models are generally organ- or task-specific rather than universal, reflecting both tumour heterogeneity and the difficulty of modelling diverse anatomical contexts. Variations in histopathology, imaging appearance, and modality physics, together with domain shifts from scanners or acquisition protocols, hinder cross-domain deployment [45, 46, 47, 48]. Public datasets are likewise organ-bound and rarely include cross-organ labels, constraining efforts toward universal detectors [49, 50]. Consequently, domain-specific systems remain standard, as they leverage anatomical priors and avoid the optimisation conflicts seen in multitask models [45, 47]. Clinical adoption also favours such models because their outputs align more closely with established diagnostic reasoning, whereas general-purpose systems often lack interpretability [46, 51].

2.4 Fusion Strategies for Multimodal Data in Medical Imaging

Multimodal DL integrates imaging, clinical, pathological, and genomic information to improve diagnostic and prognostic performance [20, 52]. Reviews consistently show that combining modalities yields measurable gains: Kline *et al.* reported a 6.4

Recent surveys highlight rapid expansion of this field: Sun *et al.* identified 77 multimodal DL studies using combinations of radiology, text, and clinical variables [53], while Haq *et al.* reported a sixteen-

fold rise in multimodal radiology publications since 2019 [54]. New reviews outline methodological trends—including integration with pathology and omics [55, 56, 57]—and show that multimodal AI is moving toward clinically aligned systems [20, 58]. This thesis focuses on DL-based fusion strategies for recognition and classification tasks, covering early, intermediate, and late fusion [52, 21], along with VLM-based approaches and loss-function considerations [20, 58, 56].

2.4.1 Early, Intermediate and Late Fusion

Early fusion. Early fusion concatenates raw or embedded features from multiple modalities into a unified representation [21, 52]. Radiomics–clinical concatenation dominated early multimodal imaging research [52], and several studies demonstrated clear improvements over unimodal models [20, 59, 60]. Despite challenges such as modality imbalance [52], early fusion remains widely used for tumour detection and risk prediction [61, 62, 63, 64].

Intermediate fusion. Intermediate (joint) fusion merges modality-specific subnetworks at a mid-level layer, enabling shared representation learning [52]. Examples include MRI–text fusion for tumour classification [65] and CT–clinical fusion for embolism severity prediction [66]. Attention and probabilistic mechanisms are common [67, 68, 69]. Transformer-based multimodal models such as IRENE and cross-attention frameworks further enhance performance [70, 71, 72]. GNN-based fusion has also emerged for integrating anatomical or genomic graphs with imaging [57]. Intermediate fusion often outperforms both early fusion and unimodal models [52].

Late fusion. Late fusion ensembles independently trained unimodal models [52]. Although it does not explicitly model cross-modal interactions, it is effective when modalities are asynchronous or incomplete. Demonstrated successes include Alzheimer's diagnosis [73], outcome prediction with image+EHR ensembles [74], and stacked ultrasound–mammography–MRI classifiers [75]. Hybrid late-fusion schemes can improve robustness under missing data [63]. It remains a transparent baseline for comparing integrated fusion strategies [52, 74].

2.4.2 VLMs and Advanced Multimodal Fusion Approaches

Recent advances in multimodal learning have been driven by VLMs and related foundation models that jointly embed images and text via large-scale self-supervised pretraining [76]. Inspired by CLIP [77], medical VLMs such as BioViL [78, 79], GLoRIA [80], MedCLIP [81, 82], and BioMedCLIP [83] align chest X-rays or other medical images with associated reports, enabling zero-shot classification and image–text retrieval. Earlier multimodal networks like TieNet had already demonstrated that incorporating clinical text improved chest X-ray classification by 5–6% AUC [84]. Surveys now catalogue over 60 medical foundation models across radiology, pathology, and ophthalmology [76], including encoder–decoder systems for report generation and visual question answering [85, 86].

A growing line of work "textualises" structured or numeric data for VLM input. AutoRad-Lung converts radiomics features into descriptive prompts for lung CT malignancy prediction [87], and FM-Bridge encodes imaging and risk variables as natural-language statements to improve liver tumour diagnosis [88]. Prompt-based methods similarly adapt general VLMs to domain-specific tasks, as in dental X-ray analysis [89]. In pathology, MIL frameworks such as MI-Zero and ViLA-MIL use CLIP-based embeddings to align gigapixel slide patches with histological text, enabling zero-shot cancer subtype classification [90, 91].

Key challenges for medical VLMs include limited domain data, specialised terminology, and factual reliability [76]. Proposed solutions span domain vocabularies, better region–word alignment, and expert feedback. MedVLM-R1 uses RLHF to refine reasoning [92], and SERPENT-VLM incorporates factuality checking to reduce hallucinations [93]. Large annotated datasets such as MIMIC-CXR and ROCO support ongoing training and transfer [20, 54]. Emerging MLLMs extend these capabilities to interactive, image–text understanding, and compact variants (e.g. Small-MM) show that clinically deployable models need not be excessively large [94, 95]. Overall, VLM-based approaches point toward holistic, patient-centred models that integrate visual, textual, and molecular information [54, 96].

Loss design in multimodal DL balances task performance with cross-modal alignment. Most models still rely on standard supervised objectives—cross-entropy for classification, Dice or pixel-wise cross-entropy for segmentation, and MSE for regression [52]. Class imbalance is commonly addressed with weighting or focal loss [58, 54], multi-label problems with sigmoid cross-entropy, and multi-task settings with weighted combinations of classification and regression terms [20]. Auxiliary modalities usually enhance representations rather than change the primary loss, although reconstruction or imputation losses can regularise models when modalities are missing [54].

Self-supervised and contrastive frameworks introduce additional objectives to enforce cross-modal consistency. InfoNCE and CLIP-style losses maximise similarity between paired image–text samples while minimising mismatched pairs [97, 76]. Medical adaptations such as BioViL and GLoRIA incorporate both global and local alignment terms, including local ranking or correlation-based losses for aligning image regions with textual phrases [80]. Reconstruction losses can further encourage one modality to predict another, regularised by L_2 penalties [21, 98]. In survival analysis, Cox partial likelihood is used to model time-to-event risk in multimodal settings [20], and hybrid tasks (e.g. segmentation plus classification) employ composite losses to balance objectives [99, 100]. Consistency- and distillation-based penalties have also been proposed for handling missing modalities and improving robustness [20, 101].

Recent models integrate more advanced objectives, including RL for multimodal generation (e.g. MedVLM-R1) [92] and joint alignment-plus-cross-entropy schemes to reduce hallucinations in VLM outputs [76]. Adversarial and distillation losses have been explored for realistic report generation and missing-modality learning [101]. Overall, the main challenge lies less in inventing new losses than

in appropriately weighting existing ones and ensuring effective gradient flow through all modality branches [52].

2.4.3 Summary

In summary, multimodal fusion strategies in medical imaging span early, intermediate, and late integration, each offering distinct trade-offs in complexity, performance, and interpretability [52, 58]. Early fusion remains simple and widely adopted, often outperforming unimodal baselines [52, 58]. Intermediate fusion, particularly attention- and transformer-based designs, enables joint feature learning and frequently delivers state-of-the-art results [52, 71, 57]. Late fusion serves as a practical, interpretable benchmark and can be advantageous when modalities are heterogeneous or asynchronously available [52, 74, 75].

A major recent development is the emergence of VLMs and multimodal foundation models, which align images, text, and textualised structured data within shared latent spaces [76, 77, 78, 80, 81, 83]. Approaches such as AutoRad-Lung and FM-Bridge suggest that converting radiomics and clinical variables into descriptive text may be a powerful strategy for leveraging VLMs in medical imaging [87, 88]. This thesis builds on these ideas by applying textualisation and VLM-based fusion to 3D renal CT, treating the VLM as a multimodal feature extractor for supervised subtype classification. Loss-function design in multimodal learning typically combines standard supervised objectives with contrastive or consistency terms to ensure effective cross-modal alignment [97, 80, 54]. Across tasks and modalities, meta-analyses show that multimodal models reliably outperform unimodal counterparts when properly optimised [58, 54, 57, 20, 52].

Persistent challenges include missing modalities, robustness and fairness across populations, and the need for interpretable models [58, 102, 54]. Explainable AI tools such as attention visualisation and prototype-based reasoning can clarify modality contributions and improve clinician trust [102, 103]. Future directions focus on unified architectures for images, text, and structured data; integration of large language models for medical reasoning; and efficient, distilled multimodal systems suitable for deployment [95, 104, 96]. More broadly, multimodal DL is evolving towards holistic, interpretable, and clinically grounded AI systems that better reflect the complexity of real-world patient care.

2.5 Interpretability and Explainability in Medical AI

In safety-critical settings such as medical imaging, interpretability—the ability to understand how a model reaches its predictions—is increasingly seen as essential for clinical adoption [105, 106]. Surveys indicate that many clinicians remain sceptical of “black-box” AI systems; for example, 71% of ICU clinicians report uncertainty about AI reliability [105]. Interpretability helps bridge this trust gap by allowing experts to verify that AI reasoning aligns with medical knowledge and

ethical standards, and it supports emerging regulatory requirements such as the EU AI Act [105]. While often conflated, interpretability differs from post hoc explainability: interpretable models are inherently transparent (e.g. linear models, decision trees) [107], whereas explainability methods (e.g. LIME, SHAP) approximate the logic of complex models without fully revealing their internal workings [107]. Recent accountable-AI frameworks advocate prioritising inherently interpretable components and using post hoc explanations as complementary tools [108, 109].

Empirical studies show that well-designed explanations improve trust calibration and decision quality [106, 110]. Heatmaps and saliency maps help radiologists assess whether a model attends to clinically relevant regions [110, 109], and pretraining on radiology-scale data can yield more meaningful visual explanations [109]. Explanations must remain human-centred—understandable, contextual, and clinically actionable [106]—especially in oncology, where opaque reasoning remains a major barrier despite high reported accuracies [111]. Interpretability also supports bias detection and error analysis before patient harm occurs [105]. Accordingly, recent work argues for explicitly trading a small amount of accuracy for greater transparency when necessary [108].

For multimodal medical AI, interpretability is increasingly viewed as a practical necessity rather than an optional add-on [109, 110]. While post hoc explainability remains useful for complex models, recent work emphasises designing transparency into model architectures from the outset [108]. This includes choosing structures amenable to explanation (e.g. attention mechanisms, case-based reasoning, generative rationales) and ensuring outputs are human-verifiable [112]. Inherently interpretable models, which combine explicit risk factors with imaging evidence in transparent ways, tend to align better with clinical reasoning and support more reliable decision-making than opaque black-box systems [107, 110, 113]. Interpretability also involves communication: dashboards, interactive visualisations, and query-based interfaces can surface AI reasoning in clinician-friendly formats [114, 112]. Achieving an appropriate balance between performance and interpretability will be crucial as multimodal AI transitions from research into routine clinical use [110, 106].

2.5.1 Interpretability across fusion strategies

Explaining multimodal models is particularly challenging because they must reveal not only which features influence predictions but also how modalities interact. Early fusion tends to entangle features from all inputs, making it difficult to attribute importance to specific modalities. In such models, explanations rely on global feature-importance or perturbation-based methods (e.g. SHAP) [112, 110], as well as occlusion experiments to quantify each modality's impact [110]. However, high-dimensional image embeddings may dominate smaller feature sets, reducing clarity and leaving these models effectively black boxes.

Intermediate fusion can improve interpretability by keeping modality-specific branches separate until mid-level integration [21]. This structure allows the use of modality-specific XAI techniques—e.g. Grad-CAM for imaging and SHAP for clinical features—followed by inspection of fusion weights or

attention maps to assess cross-modal contributions [112]. Attention-based fusion may align image regions with text tokens, more closely reflecting radiological reasoning [112]. Gating mechanisms that assign instance-specific modality weights offer additional transparency [112]. Nonetheless, attention weights do not automatically guarantee faithful explanations and require validation [109, 114].

Late fusion, which combines separate unimodal predictions, offers the greatest modality-wise transparency but limited insight into interactions. Clinicians can interpret each branch independently, using heatmaps for imaging and feature charts for clinical data, mirroring multidisciplinary reasoning [105]. Ensemble-level XAI methods can approximate cross-modal effects [112]. While late fusion may sacrifice some representational synergy, its modularity and clarity make it attractive in safety-critical workflows [105].

2.5.2 Vision–language and foundation models

VLMs naturally support more intuitive explanations by operating directly on images and text. Rather than exposing only abstract feature vectors, they can generate natural-language justifications for predictions, aligning AI reasoning with radiological narratives [112, 114]. Attention maps in CLIP-style models reveal associations between image regions and semantic tokens, while architectures such as GLoRIA explicitly link patches to descriptive words [109, 80]. Textualisation of non-image inputs (e.g. lab values, genomics, radiomics) further enhances interpretability by mapping heterogeneous data into a shared language space [112]. However, VLMs can hallucinate or overgeneralise, motivating the use of factuality constraints and RLHF to ensure clinically reliable explanations [114]. As multimodal foundation models mature, they hold promise for systems that combine strong predictive performance with human-readable, clinically meaningful explanations.

2.5.3 Clinician acceptance and evaluation of XAI

Clinician acceptance is central to effective interpretability. Studies indicate that personalised, case-specific explanations improve trust and support appropriate reliance on AI recommendations [110]. Vani *et al.* showed that SHAP- and attention-based visualisations enhanced physicians' understanding of model outputs [110], while other work argues that explainable AI can increase guideline adherence when explanations are clear and contextually relevant [105, 107]. Poor or misleading explanations, by contrast, may reduce safety and confidence [105, 109]. Consequently, a "clinician-in-the-loop" approach is increasingly advocated, where experts evaluate AI explanations through user studies comparing diagnostic performance with and without explanations [109, 110]. Standards for clinical explainability, such as understandability, truthfulness, and actionability, are being proposed [114, 115], and professional societies and regulators now highlight interpretability as a key requirement for future AI approval [105, 107].

2.6 Research Gaps

First, renal CT imaging research still relies predominantly on handcrafted radiomics with conventional ML classifiers, with limited use of DL and few controlled comparisons between the two paradigms. Reported performances vary widely owing to differences in imaging phase, feature extraction, and evaluation criteria, leading to poor reproducibility and generalisability.

Second, although high-quality public datasets such as KiTS23 and TCGA/TCIA exist, they are not yet systematically exploited, and external validation remains uncommon. Standardised pipelines and benchmark evaluations on public data are needed to enable robust comparison across studies.

Third, while prior work highlights the potential of early, intermediate, and late fusion, as well as transformer-based and VLM approaches, renal tumour imaging has seen few systematic comparisons of these multimodal strategies or analyses of how fusion mechanisms affect accuracy, robustness, and interpretability. In particular, textualising radiomics features and aligning them with images within a VLM framework remains largely unexplored, despite its promise for improving clinical transparency and human-understandable reasoning.

Fourth, most existing studies omit patient-level metadata that could enrich the multimodal context, and rarely quantify robustness to missing modalities or interpretability aligned with radiologist reasoning. Without explicit assessment of how models communicate their decision rationale, the translational value and clinical trustworthiness of multimodal AI remain uncertain.

Finally, although recent work has proposed various multimodal loss functions, task-specific strategies for balancing supervised objectives with cross-modal alignment and consistency losses in 3D renal CT classification are still underexplored.

In summary, this thesis addresses these gaps by: (1) establishing reproducible baselines with external validation on public 3D CT datasets; (2) systematically comparing multiple fusion strategies, including transformer-based and textualised VLM-based integration of radiomics; (3) incorporating patient metadata to build tri-modal models; and (4) analysing loss-function designs that balance supervised and cross-modal learning. Together, these directions aim to improve fusion efficacy, robustness, and interpretability, and to support clinically trustworthy, explainable multimodal AI for renal tumour analysis.

Based on the research gaps explored in the literature review, this thesis aims to answer the research questions mentioned in section 1.3.

CHAPTER

THREE

METHODOLOGY AND RESULTS

3.1 Introduction

This chapter presents the methodological progress made to date, focusing primarily on evaluating whether DL can feasibly perform texture-based tumour classification, thereby addressing RQ1. Section 3.2 establishes a radiomics-only baseline on the KiTS23 dataset, while Section 3.3 validates the technical feasibility of integrating radiomics with deep features in a lightweight 2D setting. Section 3.4 extends these fusion strategies to 3D CT data, providing further evidence for RQ1 and laying essential groundwork for RQ2 by exploring how different fusion mechanisms may influence model performance. These studies collectively form the empirical basis for the more advanced multimodal fusion investigations planned in subsequent work.

3.2 Renal Cell Carcinoma Texture Analysis with KiTS23 Dataset (under review as a journal article)

To address RQ1, "How can radiomics and DL representations be effectively integrated to improve texture-based classification performance across medical imaging modalities?", an appropriate dataset is selected to perform texture analysis using a combination of traditional ML and radiomics features. The results of this approach serve as a benchmark for subsequent comparisons with DL and feature fusion methods.

3.2.1 Dataset

The KiTS23 dataset is part of the Kidney Tumour Segmentation Challenge 2023 [34]. The detail of the dataset is introduced in section 2.1.3.

3.2.2 Results

This section presents the ML models integrated with RFE used for statistical analysis in this study and provides a comparative summary of their performance. The objective of the statistical analysis is to distinguish between ccRCC and other types of renal tumours. In the dataset, the number of ccRCC cases and other types of tumours is balanced, with an approximate ratio of 6.5:3.5.

The ML models employed for classification include Random Forest, Support Vector Classifer (SVC), Gradient Boosting, and Logistic Regression. Among the four models, the Logistic Regression exhibited significantly better performance compared to the other three models. Specific comparisons of their performances are detailed in the evaluation subsection.

3.2.2.1 Quantitative Evaluation

Table 3.1: Evaluation scores of the classification for the four benchmarking methods: Random Forest, SVC, Gradient Boosting, and Logistic Regression

Metric (%)	Random Forest	SVC	Gradient Boosting	Logistic Regression
Accuracy	74.1 ± 2.45	71.2 ± 2.65	70.0 ± 3.8	**75.1** ± 4.4
AUC	77.9 ± 4.85	77.5 ± 3.65	73.8 ± 6.1	**79.6** ± 5.0
Precision	78.0 ± 3.9	74.2 ± 2.5	74.2 ± 3.0	**78.1** ± 3.0
Sensitivity	83.9 ± 3.85	84.6 ± 3.95	81.8 ± 2.5	**85.0** ± 5.3
Specificity	56.7 ± 10.95	47.1 ± 8.35	49.0 ± 7.7	**57.3** ± 6.8
Number of Features	14	10	16	41

In this study, five evaluation metrics were utilised: accuracy, Area Under the Curve (AUC), precision, sensitivity, and specificity. The 0.95 confidence intervals for each metric of every model were also calculated, and all performance metrics were obtained using five-fold cross-validation. During the recursive feature elimination (RFE) process, model performance was tracked as the number of selected features varied. The SVC model achieved its highest AUC when RFE retained only 10 features, whereas the Logistic Regression model reached its peak AUC with a broader set of 41 features. Please refer to Table 3.1 for the number of features corresponding to the optimal AUC for each model, together with their performance across the other evaluation metrics.

A comparison of model performances in Table 3.1 clearly shows that the Logistic Regression classifier excels in identifying clear cell renal cell carcinoma (ccRCC), achieving the highest scores across all five metrics, with an AUC of 0.796.

3.3 Fusing Radiomics Features with Deep Representations for Gestational Age Estimation in Fetal Ultrasound Images (Accepted at MICCAI 2025)

To address RQ1, which concerns the integration of radiomics and DL representations for improved texture-based classification, an initial feasibility study was conducted on gestational age (GA) estimation using lightweight 2D fetal ultrasound images. This provided a controlled setting for evaluating fusion strategies before extending the methodology to 3D CT-based renal tumour analysis.

This work introduces a feature-fusion framework that integrates radiomics features with CNN-derived deep representations of the fetal head to improve GA prediction. Two fetal ultrasound datasets were used to assess the approach, and both ML and DL baselines were included for comparison. The main contributions are: (i) proposing the first framework that fuses radiomics and ultrasound images for GA estimation; (ii) introducing a plug-and-play cross-attention module for effective fusion of radiomics and deep features; and (iii) demonstrating, through comprehensive experiments, the superiority of the fusion module over unimodal ML and DL models.

3.3.1 Datasets

Two publicly available datasets were employed in this study. The Spanish trans-thalamic dataset (ES-TT) originates from two centres in Spain [116], while the HC18 dataset was sourced from a database in the Netherlands [117]. All scans were acquired from healthy singleton pregnancies. The HC18 data were collected using two ultrasound machines at a single centre, whereas ES-TT data were obtained from six machines across two different centres. Owing to the variability of the fetal head morphology across different pregnancy trimesters, an additional challenge arises in the estimation of GA. The two datasets were combined and partitioned in a 70:30 ratio for training and testing. From the test set, a subset of 100 images was randomly selected for validation purposes.

3.3.2 Methodology

An overview of the fusion framework is shown in Fig. 2. It consists of three key components: (1) a pre-trained ConvNeXt network [118] for learning deep representations; (2) a Python-based pipeline for radiomics feature extraction; and (3) a cross-attention module that fuses radiomics and deep features for GA prediction.

The proposed framework employs the CNN topology to construct the DL model to learn deep representations that reflect high-level semantic information of ultrasound images. Specifically, the study incorporate models ResNet18 [119], SwinTransformer [120], EfficientNet V2 [121], MobileNet V3 [122], ViT [123], MaxViT [124], and ConvNeXt [118] for their proven effectiveness in image clas-

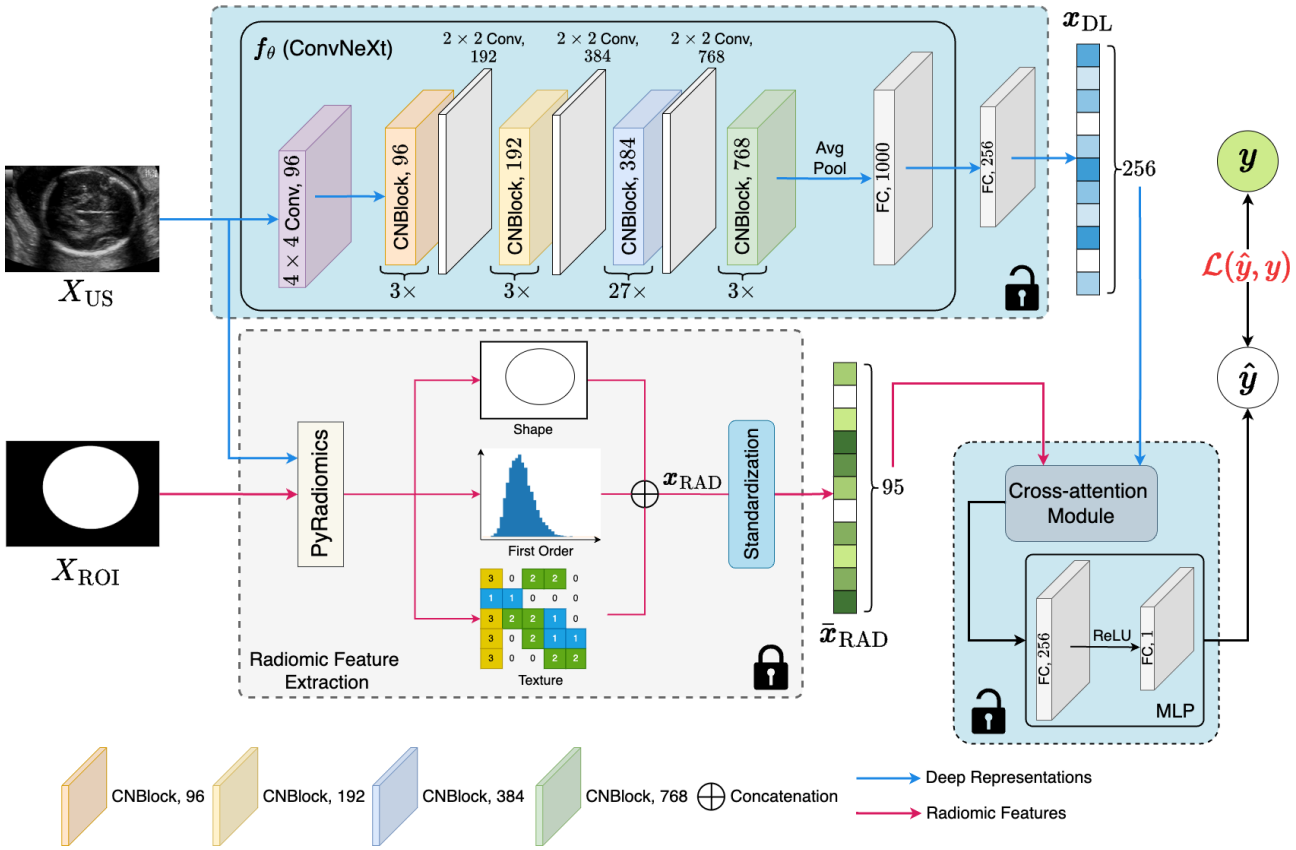

Figure 2: Proposed fusion framework for GA estimation from fetal ultrasound images. *Blue*: Trainable (DL model, cross-attention, MLP). *Gray*: Non-trainable (Radiomics)

sification tasks. The DL model in the proposed framework adopts the ConvNeXt stack architecture. Features are passed to a cross-attention module to perform a fusion operation.

3.3.3 Results

GA prediction accuracy was evaluated using MAE and RMSE, which were highly correlated ($r > 0.998$); therefore, only MAE is reported. For ML models, the coefficient of determination (R^2) was additionally used to assess model fit. As shown in Table 3.2, the proposed framework achieved the best performance with an average MAE of 8.0 days. The cross-attention module significantly improved GA estimation compared with both the concatenation module ($p < 0.001$) and the baseline model ($p < 0.001$). Although ResNet18 and EfficientNet V2 showed slightly higher MAE under cross-attention than under concatenation, overall performance across all architectures was still significantly better with cross-attention ($p < 0.001$), demonstrating the advantage of the fusion mechanism.

Table 3.3 summarises the performance of radiomics-based ML models. While GBR with RFE achieved the highest R^2, its MAE remained substantially higher than that of the proposed framework. Overall, the fusion approach consistently outperformed all ML baselines.

Attention Weight Visualisation. Fig. 3 shows cross-attention maps from representative cases across

Table 3.2: Test results of GA estimation by integrating DL models with concatenation and cross-attention mechanisms are compared. C: Concatenation. XA: Cross-attention.

DL Model	C	XA	MAE ↓	P-value
ResNet18	✗	✗	9.9	
EfficientNet V2	✗	✗	9.3	
MaxViT	✗	✗	10.7	$<10^{-3}$
SwinTransformer	✗	✗	9.4	
ConvNeXt	✗	✗	8.6	
ResNet18	✓	✗	10.2	
EfficientNet V2	✓	✗	8.6	
MaxViT	✓	✗	9.6	$<10^{-3}$
SwinTransformer	✓	✗	8.6	
ResNet18	✗	✓	10.4	
EfficientNet V2	✗	✓	8.6	
MaxViT	✗	✓	8.7	–
SwinTransformer	✗	✓	8.3	
ConvNeXt	✗	✓	**8.0**	

Table 3.3: Test results of GA estimation using ML models by employing two techniques to select radiomics features. SVM: Support Vector Machine. KNN: K-Nearest Neighbors. RF: Random Forest. GBR: Gradient Boosting Regression.

ML Model	MAE ↓	R^2 ↑
SVM (LASSO)	25.2	0.62
KNN (LASSO)	25.2	0.61
AdaBoost (LASSO)	26.8	0.60
Ridge (LASSO)	25.7	0.64
RF (LASSO)	21.7	0.69
GBR (LASSO)	21.8	0.69
SVM (RFE)	24.9	0.62
KNN (RFE)	24.5	0.62
AdaBoost (RFE)	26.5	0.61
Ridge (RFE)	25.8	0.64
RF (RFE)	21.4	0.69
GBR (RFE)	**21.3**	**0.70**

Table 3.4: Ablation study on the different feature fusion modules and pre-trained weights. Concat: Concatenation. XA: Cross-attention.

Model	Image	Pretrain	Concat	Cross-attention	MAE ↓
ConvNeXt	✓	✗	✗	✗	28.2
ConvNeXt	✓	✓	✗	✗	8.6
ConvNeXt + Concat	✓	✓	✓	✗	8.7
ConvNeXt + XA (Ours)	✓	✓	✗	✓	**8.0**

the three trimesters. The module highlights complementary patterns between radiomics features and deep representations, particularly in the second and third trimesters. In early-trimester data, attention weights emphasise specific radiomics features such as Inverse Difference and Inverse Difference Moment. These visualisations provide an interpretable indication of how individual radiomics features contribute to GA estimation.

Ablation Study. The contributions of fine-tuning, fusion strategy, and cross-attention were assessed through an ablation study. As shown in Table 3.4, fine-tuning with pre-trained weights is essential for good performance, reducing MAE from 28.2 to 8.6 days. Simple concatenation offers minimal additional benefit, whereas incorporating cross-attention further improves accuracy to 8.0 days, confirming its importance for effective fusion of deep and radiomics features.

3.3.4 Conclusion

This study presents a feature fusion framework that integrates image-based deep representations with radiomics features to estimate fetal GA from ultrasound images. Evaluated on a lightweight dataset and a simple prediction task, the approach enabled rapid prototyping while still achieving strong predictive

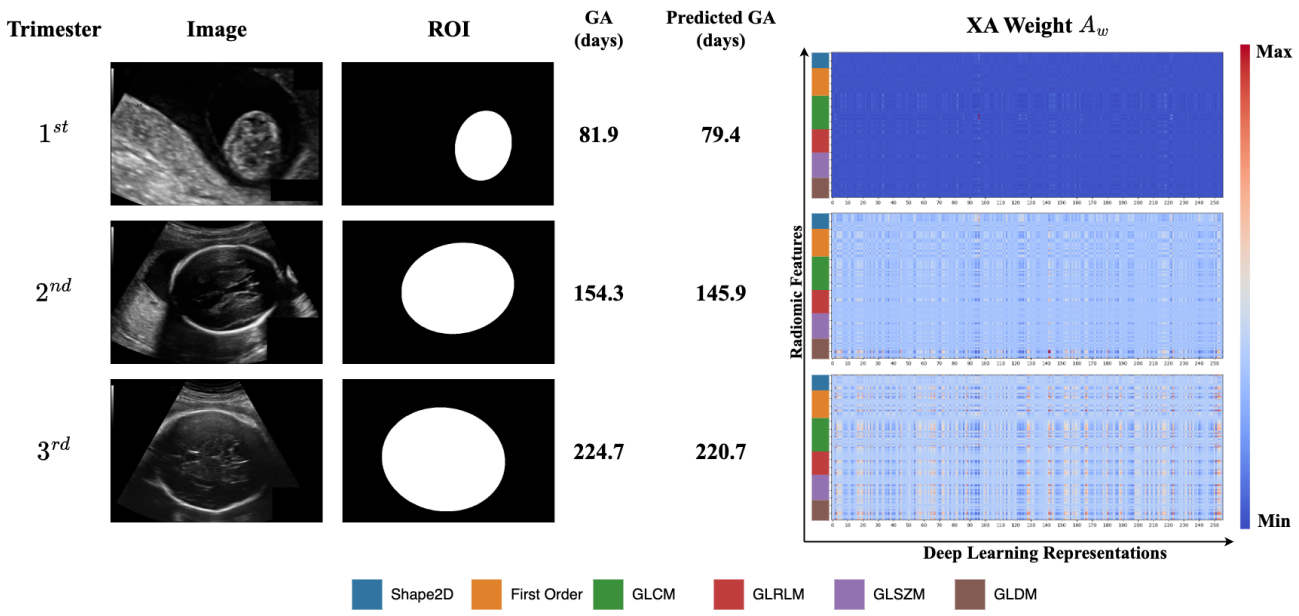

Figure 3: Visualisation of attention weights within our framework.

performance. These results highlight the value of combining handcrafted and learned features. The successful 2D proof of concept provides a methodological basis for extending the fusion strategy to more complex 3D CT-based renal tumour classification, where multimodal feature integration is likely to have even greater impact.

3.4 Fusion-Based DL Models for 3D Renal CT Classification

The experiments presented in this section constitute a central component of the investigation and directly address RQ1, which concerns the effective integration of radiomics and DL representations for improved texture-based classification performance across medical imaging modalities. They also resolve a substantial portion of RQ2 and RQ3, which focus on the identification of the most effective feature-fusion strategies for aligning radiomics features with imaging representations, including both conventional numerical fusion and more advanced approaches that will later involve VLMs. The purpose of the present section is to establish a unified and controlled benchmark that compares radiomics-only ML models, image-only DL models, and a series of multimodal fusion architectures under consistent experimental conditions.

3.4.1 Methods

The binary classification task remains the distinction between ccRCC and non-ccRCC tumours as defined earlier in section 3.2. All experiments were conducted on the KiTS23 dataset, whose multi-centre origin and varied scanner configurations provide substantial heterogeneity that promotes generalisability. The TCGA dataset is currently being prepared for use as an external test set. A three-way

split of approximately 60% training, 25% validation, and 15% testing was adopted for all models to ensure consistent evaluation. For the radiomics-only ML models, the training and validation partitions were pooled to create a larger training set while preserving the identical test set used for all modelling paradigms. The details of the KiTS23 dataset are in section 2.1.3.

A strengthened preprocessing workflow based on MONAI was applied across all experiments [125]. Very small tumours and multi-lesion cases were excluded to avoid unstable segmentation boundaries. Multiple expert annotations were fused using STAPLE in order to obtain consensus tumour masks. Each mask was expanded by twelve voxels in all directions and then cropped to produce standardised volumetric regions of interest suitable for 3D convolutional networks. Radiomics features were extracted from both the original and the Laplacian of Gaussian filtered images. Feature selection was then performed in multiple stages to retain only stable and non-redundant radiomics predictors. A sequence of robustness filtering steps, including ICC, MAD, Pearson correlation, and clustering, reduced the initial 386 features to 71 stable descriptors. LASSO was subsequently applied to derive a compact 5-feature subset used for radiomics-only ML. These five features constitute a highly-reduced radiomics subset that was also evaluated in later fusion experiments

The radiomics-only models were trained using the five LASSO-selected features to establish a compact and interpretable baseline that reduces the risk of overfitting while maintaining discriminatory potential. Several classifiers were evaluated, including Logistic Regression, Gradient Boosting, Random Forests, XGBoost, and SVM with an RBF kernel.

The image-only DL baselines were constructed using a 3D ResNet18 architecture. One model was trained from scratch, while the other was initialised with MedicalNet weights. Both models were trained on the cropped 3D regions of interest using weighted cross-entropy to compensate for the slight class imbalance present in KiTS23.

A series of feature fusion models was then developed to integrate radiomics features with the deep embeddings produced by the 3D ResNet18 encoder. Five fusion mechanisms were examined. The first was simple concatenation, which served as an early-fusion baseline. The second and third were two forms of cross-attention, the first using a shared linear transformation for all radiomics features and the second introducing feature index embeddings to preserve structural information. The fourth model employed gated attention in which radiomics and deep representations were adaptively weighted. The fifth model used an SE-like recalibration mechanism that amplified or suppressed radiomics contributions according to the corresponding deep features. All models were trained and evaluated using identical splits, and their final test performance was assessed primarily using the AUC, with specificity reported for contextual interpretation. All results correspond to those presented in Table 3.5.

Table 3.5: Performance of radiomics-only, DL and fusion models. All metrics are reported as percentages. Fusion-65 models combine image features with all 65 radiomics features, while Fusion-3 models fuse image features with the reduced 3-feature subset.

Model	AUC	ACC	F1	Precision	Recall	Specificity
ML (Radiomics-only)						
GradientBoosting	69.4	71.7	78.8	82.1	76.1	61.2
LightGBM	66.8	66.8	74.1	80.7	69.0	61.3
LogisticRegression	73.5	68.9	74.8	85.3	67.0	73.2
RandomForest	69.7	65.5	70.4	86.6	59.7	78.6
SVM_RBF	74.4	62.0	65.6	**88.3**	52.6	**83.8**
XGBoost	67.0	65.4	71.9	82.3	64.3	67.8
DL						
3D ResNet18	52.1	63.3	75.0	71.7	78.6	27.8
3D ResNet18 (MedicalNet pre-trained)	71.0	61.7	72.9	72.1	73.8	33.3
Fusion-65						
Concat	74.9	**83.3**	**88.4**	86.4	**90.5**	66.7
Cross Attention (shared Linear)	56.5	58.3	67.5	74.3	61.9	50.0
Cross Attention V2 (with feature embedding)	70.2	65.0	71.2	83.9	61.9	72.2
Gated Attention	**81.2**	76.7	83.3	83.3	83.3	61.1
SE-like Radiomics Recalibration	67.3	65.0	73.4	78.4	69.0	55.6
Fusion-3						
Concat	70.0	66.7	75.6	77.5	73.8	50.0
Cross Attention (shared Linear)	70.9	56.7	61.8	80.8	50.0	72.2
Cross Attention V2 (with feature embedding)	73.3	75.0	81.5	84.6	78.6	66.7
Gated Attention	79.1	81.7	86.7	87.8	85.7	72.2
SE-like Radiomics Recalibration	68.4	63.3	73.8	73.8	73.8	38.9

3.4.2 Results

The radiomics-only ML models showed that a small number of well-selected features can capture meaningful tumour heterogeneity. SVM achieved the strongest baseline (AUC = 0.744), followed by Logistic Regression (AUC = 0.735), while Gradient Boosting, Random Forest, and XGBoost produced AUCs of 0.668–0.697. Specificity remained stable across models, confirming that the Youden-index thresholding provided balanced sensitivity–specificity trade-offs.

Image-only DL models performed notably worse: 3D ResNet18 achieved AUCs of 0.521 (from scratch) and 0.544 (MedicalNet pretraining), with similarly low specificity. These results indicate that volumetric CNNs struggle to learn discriminative renal texture patterns from moderate-sized datasets, and that pretraining offers only marginal benefit.

Multimodal fusion consistently outperformed both unimodal paradigms. Simple concatenation achieved an AUC of 0.749, surpassing all radiomics-only and image-only baselines. The first cross-

attention variant, which used a shared linear transformation across radiomics features, performed poorly (AUC = 0.565), suggesting that collapsing individual feature identities limits effective interaction. The refined cross-attention version with feature-index embeddings improved performance to 0.702 and provided a more stable sensitivity–specificity relationship.

Gated attention was the best-performing approach. With the full 71-feature set, it achieved an AUC of 0.812—the highest among all models [126]—and maintained balanced accuracy and specificity. Even with only 5 radiomics features, it achieved an AUC of 0.791, indicating strong robustness and effective modality-weighting. By adaptively modulating radiomics and deep CT representations, gated attention mitigates heterogeneous feature scales and suppresses modality-specific noise, enhancing generalisation.

The SE-like recalibration module achieved an AUC of 0.673 [127], outperforming DL-only baselines but remaining below concatenation and gated-attention fusion. Some fusion variants provided only small gains or slight declines, likely due to suboptimal modality alignment or overfitting, indicating that simple feature merging is insufficient without an interaction mechanism.

Fusion models also enable enhanced interpretability. Attention, gating, and recalibration modules yield explicit feature-weighting patterns that map to radiomics descriptors or spatial CT features. These provide clinically meaningful insight into which quantitative attributes or tumour regions influence predictions—capabilities not available in image-only models. Thus, radiomics–DL fusion offers a transparent cross-modal reasoning framework that supports clinical validation.

Overall, multimodal fusion markedly improves renal tumour texture classification. Radiomics-only models remain strong and interpretable, but DL alone is inadequate for this task. Fusion approaches consistently surpassed unimodal baselines, with gated attention offering the most robust and generalisable performance. These findings directly answer RQ1 and substantially address RQ2 by demonstrating the benefit of integrating radiomics with DL representations and identifying which fusion mechanisms are most effective. The benchmarks established here form the foundation for subsequent experiments involving vision–language fusion and incorporation of clinical metadata, supporting future development of richer multimodal renal tumour classification systems.

CHAPTER

FOUR

RESEARCH PLAN

4.1 Future Work Plan

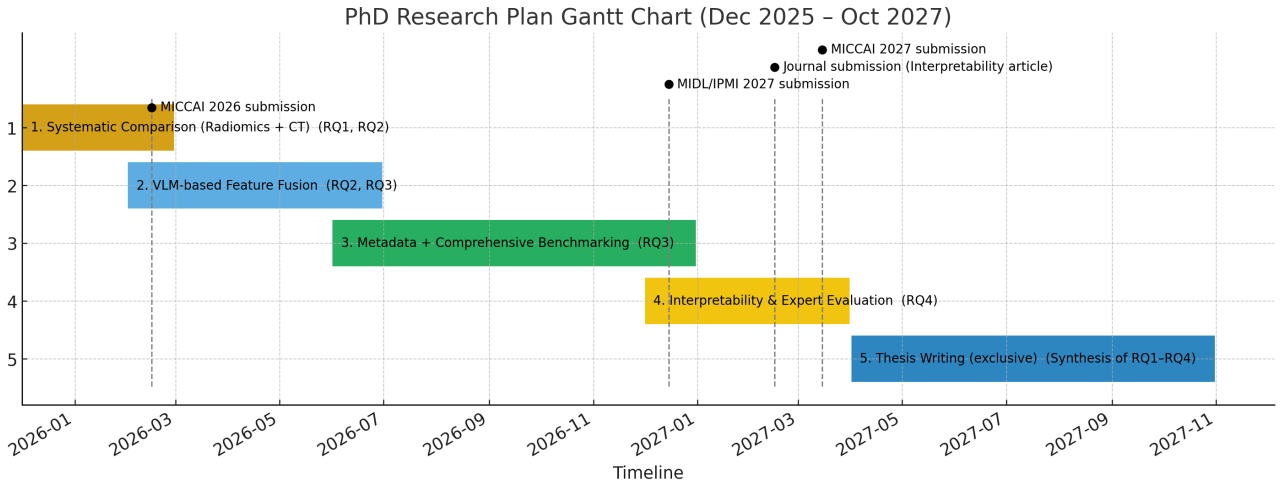

Figure 4: Gantt chart of the planned PhD research activities (December 2025 – October 2027). Research tasks (1–4) address specific research questions (RQs) and conclude by March 2027, followed by a dedicated thesis-writing phase (April–October 2027). Key milestones include submissions to **MICCAI 2026**, **MIDL/IPMI 2027**, the **Interpretability Journal Article**, and **MICCAI 2027**.

The forthcoming research period, from December 2025 to October 2027, will concentrate on advancing multimodal fusion and interpretability methods for 3D CT-based renal tumour classification. A visual summary of the timeline and planned milestones is shown in Figure 4. The external validation dataset (TCIA–KIRC) is already available and will be consistently used to ensure reproducibility and generalisability. Since the adaptation of the 2D fusion framework to 3D CT has been completed, the remaining work focuses on comparing fusion strategies, developing advanced vision–language and metadata-based fusion models, and performing interpretability studies. All experimental work will conclude by March 2027, reserving April – October 2027 exclusively for thesis writing. A summary of the planned research tasks, their expected outputs, and the corresponding research questions is presented in Table 4.1.

1. Systematic Comparison of Multimodal Fusion Strategies (Radiomics + CT) (December 2025

Table 4.1: Summary of planned research tasks, outputs, and their corresponding research questions (RQs).

Task	Focus / Title	Timeline	Planned Output / Venue	Addressed RQs
1	Systematic Comparison of Multimodal Fusion Strategies (Radiomics + CT)	Dec 2025 – Feb 2026	MICCAI 2026 (Full paper submission Feb 2026)	RQ1, RQ2
2	VLM-based Feature Fusion	Feb 2026 – Jun 2026	MIDL 2027 or IPMI 2027	RQ2, RQ3
3	Metadata Integration and Comprehensive Benchmarking	Jun 2026 – Dec 2026	MICCAI 2027 (Submission Mar 2027)	RQ3
4	Interpretability and Expert Evaluation	Dec 2026 – Mar 2027	Journal submission on model interpretability	RQ4
5	Thesis Writing and Submission	Mar 2027 – Oct 2027	PhD Thesis	

– February 2026)

This phase will perform a comprehensive comparison of multimodal fusion architectures combining radiomics and deep CT representations. Several paradigms, including early, attention-based intermediate, and late fusion, will be evaluated using the KiTS23 dataset and externally validated on TCIA–KIRC. Model architectures will be standardised with consistent training protocols to ensure comparability, and statistical significance testing (e.g., paired t-tests, DeLong's test) will be applied to evaluate performance differences across fusion methods. Quantitative performance will be measured using metrics such as AUC, accuracy, sensitivity, specificity, and cross-dataset generalisability, while calibration and uncertainty estimation will also be examined to assess model reliability. The findings will be prepared for submission to **MICCAI 2026** (full paper deadline February 2026; conference October 2026), representing the first major dissemination of the multimodal framework results.

This task directly addresses RQ1 and RQ2 by identifying optimal multimodal fusion strategies that most effectively combine radiomic and deep visual representations for renal tumour classification.

2. VLM-based Feature Fusion (February 2026 – June 2026)

Building on the comparative analysis, this task will investigate advanced VLM-based multimodal fusion. Radiomic features will be converted into descriptive textual prompts capturing tumour morphology, intensity, and heterogeneity, and these will be co-encoded with volumetric CT data using pretrained medical VLMs such as BioMedCLIP and MedCLIP. Further experiments will examine the effects of prompt engineering and fine-tuning strategies on cross-modal representation learning. The performance of VLM-driven fusion will be benchmarked against conventional numerical fusion,

supported by quantitative and qualitative evaluations to assess interpretability and clinical coherence. The resulting work is intended for submission to **MIDL 2027** or **IPMI 2027**.

This task directly addresses RQ2 and RQ3 by developing semantically aligned multimodal representations that improve model robustness and interpretability across imaging modalities.

3. Metadata Integration and Comprehensive Benchmarking (June 2026 – December 2026)
This stage will incorporate patient metadata into the multimodal fusion framework to enrich contextual understanding and improve predictive performance. Different integration strategies will be examined. In one approach, metadata will be embedded using tabular encoders and fused with the joint radiomic–CT representation via concatenation or attention-based mechanisms. Alternatively, selected metadata variables will be converted into natural-language descriptions (e.g., "a 65-year-old male patient with a left-sided tumour") and injected into the vision–language fusion pipeline to evaluate whether semantically aligned metadata improves interpretability and decision consistency. Cross-validation and ablation studies will quantify the individual contribution of each information source.

Benchmarking will compare five model classes: a radiomics-only ML baseline; a CT-only DL model; a radiomics–CT fusion model; a VLM-based multimodal fusion framework; and a tri-modal system integrating radiomics, CT imaging, and patient metadata. This comprehensive comparison will provide insights into the relative advantages of classical, unimodal, and multimodal approaches. Results will be prepared for a **MICCAI 2027** submission (estimated deadline March 2027; conference October 2027), forming a key milestone for the multimodal learning component of the project.

This task directly addresses RQ3 by assessing how the inclusion and semantic encoding of metadata enhance model accuracy, contextual understanding, and generalisability within a unified benchmarking framework.

4. Interpretability and Expert Evaluation (December 2026 – March 2027)
Before thesis writing commences, the interpretability of the best-performing multimodal models will be rigorously examined. Rather than relying primarily on post-hoc explanation tools, interpretability will be achieved by visualising the contribution and influence of radiomic and metadata features within the decision-making process. Feature attribution will be evaluated intrinsically through the model's fusion attention maps, weighting coefficients, and latent-space activations that reflect the importance of different modalities. For metadata-augmented and VLM-based models, visualisation will focus on how contextual variables or textual descriptions guide the final predictions. Representative cases will be qualitatively assessed by expert radiologists to evaluate the clinical plausibility of the model's reasoning and the correspondence between salient features and radiological criteria. The outcomes of this phase will form the basis of a journal submission dedicated to interpretable and clinically transparent multimodal AI for renal imaging.

This task directly addresses RQ4 by investigating the intrinsic interpretability of multimodal fusion models and validating whether the identified radiomics and metadata contributions align with expert

radiological reasoning.

5. Thesis Writing (March 2027 – October 2027)
The final six months will be dedicated to thesis preparation and submission. All experimental findings, benchmarking results, and interpretability analyses will be synthesised into a cohesive dissertation. Writing efforts will focus on integrating the theoretical framework, methodological developments, and empirical evidence into a unified narrative, with supervisory feedback and internal review used to refine clarity, structure, and academic quality. This phase consolidates the outcomes of RQ1–RQ4 into a coherent doctoral thesis that articulates the theoretical and methodological contributions of the project.

4.2 Future Vision: Towards Immersive and Interpretable Clinical AI

If RQs 1–4 are successfully addressed, this PhD will deliver a reproducible and interpretable multimodal AI framework for renal tumour analysis. Beyond improving algorithmic performance, the broader ambition of this work is to reshape how clinicians perceive, interact with, and understand complex medical data.

Immersive technologies such as virtual reality (VR) represent a natural extension of multimodal AI. By transforming volumetric CT data and radiomics-informed segmentations into interactive 3D representations, VR can offer clinicians an intuitive spatial understanding of tumour morphology and its anatomical context [128]. Moving from conventional 2D slices to immersive 3D exploration can enhance surgical planning, education, and multidisciplinary collaboration, effectively bridging quantitative AI predictions with human spatial reasoning.

The interpretability agenda of this thesis, linking radiomics features, deep representations, and patient metadata, aligns closely with such immersive environments. VR platforms could dynamically highlight radiomic signatures, visualise texture heterogeneity, and display AI explanations directly within the 3D tumour model. This would allow clinicians not only to observe tumour structure but also to understand the rationale behind AI-driven classifications in real time.

Achieving the RQs will therefore provide the technical and conceptual foundation for an immersive clinical AI system. A subsequent postdoctoral programme could integrate the developed multimodal models into VR interfaces, enabling interactive interpretability and shifting the contribution of this research from analytical modelling to experiential clinical insight.

In essence, this work lays the groundwork for a future in which multimodal AI and VR coexist within a unified clinical interface. Such an environment could transform decision-making, medical training, and patient communication by allowing human expertise and machine intelligence to converge in the same immersive space.

CHAPTER

FIVE

CONCLUSION

This report summarises the completed progress and outlines the remaining trajectory of a PhD project dedicated to advancing multimodal AI for renal tumour analysis. Through an extensive literature review and systematic evaluation, the research confirmed the promise of DL for medical texture analysis while identifying key shortcomings, particularly the underuse of DL in this domain and the absence of systematic fusion frameworks combining radiomics with deep features.

To address these challenges, a feature fusion framework was proposed and validated on a 2D task, demonstrating the feasibility and interpretability advantages of integrating handcrafted and learned representations. This early validation provided a robust foundation for its extension to 3D CT-based renal tumour classification.

The forthcoming research will expand this framework to large-scale volumetric datasets (KiTS23 and TCGA–KIRC), conduct benchmarking across radiomics-only, CT-only, and multimodal fusion models, and incorporate metadata using both tabular and language-based encodings. Vision–language models will be explored for cross-modal integration, leveraging their capability to translate radiomic features into semantic descriptions that align with radiological reasoning. Interpretability will be achieved through the visualisation of radiomic and metadata contributions, enabling direct clinical insights rather than relying solely on post hoc explanation tools.

This project contributes to bridging the methodological and translational gaps in medical AI by developing a reproducible, interpretable, and clinically relevant multimodal framework for renal tumour classification. With a clearly defined research roadmap, validated methodology, and ongoing dissemination through peer-reviewed venues, the project is positioned to deliver meaningful academic and clinical impact in the field of AI-assisted oncology.

My confidence in completing this research is grounded in prior academic and practical experience. I hold a master's degree with three years of research in computer vision. I ranked first in the ML-Labs Bootcamp, attended three competitive AI summer schools, and was the only UCD computer science student selected for the UNA Europa Challenge 2025. I also published in the best conference in my field. These experiences have strengthened my technical capabilities and interdisciplinary outlook, ensuring that I am well-prepared to complete this research successfully.

CHAPTER

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BIBLIOGRAPHY

[1] Hyuna Sung, Jacques Ferlay, Rebecca L. Siegel, Mathieu Laversanne, Isabelle Soerjomataram, Ahmedin Jemal, and Freddie Bray. Global cancer statistics 2020: Globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, 71(3):209–249, 2021.

[2] Laura Bukavina, Karim Bensalah, Freddie Bray, Maria Carlo, Ben Challacombe, Jose A. Karam, Wassim Kassouf, Thomas Mitchell, Rodolfo Montironi, Tim O'Brien, Valeria Panebianco, Ghislaine Scelo, Brian Shuch, Hein van Poppel, Christopher D. Blosser, and Sarah P. Psutka. Epidemiology of renal cell carcinoma: 2022 update. *European Urology*, 82(5):529–542, 2022.

[3] James Hsieh, Mark Purdue, Sabina Signoretti, Charles Swanton, Laurence Albiges, Manuela Schmidinger, Daniel Heng, James Larkin, and Vincenzo Ficarra. Renal cell carcinoma. *Nature Reviews Disease Primers*, 3:17009, 03 2017.

[4] Brian I Rini, Steven C Campbell, and Bernard Escudier. Renal cell carcinoma. *The Lancet*, 373(9669):1119–1132, 2009.

[5] Amparo Sánchez-Gastaldo, Emmanuelle Kempf, Aránzazu González del Alba, and Ignacio Duran. Systemic treatment of renal cell cancer: A comprehensive review. *Cancer Treatment Reviews*, 60:77–89, 2017.

[6] Nadezda Petejova and Arnost Martinek. Renal cell carcinoma: Review of etiology, pathophysiology and risk factors. *Biomedical papers of the Medical Faculty of the University Palacky, Olomouc, Czechoslovakia*, 160, 11 2015.

[7] Gavin Low, Guan Huang, Winnie Fu, Zaahir Moloo, and Safwat Girgis. Review of renal cell carcinoma and its common subtypes in radiology. *World Journal of Radiology*, 8:484, 05 2016.

[8] Richard Gray and Gabriel Harris. Renal cell carcinoma: Diagnosis and management. *American family physician*, 99:179–184, 02 2019.

[9] Julia Mühlbauer, Luisa Egen, Karl-Friedrich Kowalewski, Maurizio Grilli, Margarete Walach, Niklas Westhoff, Philipp Nuhn, Fabian Laqua, Bettina Baeßler, and Maximilian Kriegmair. Radiomics in renal cell carcinoma-a systematic review and meta-analysis. *Cancers*, 03 2021.

[10] Daniel Sullivan, Nancy Obuchowski, Larry Kessler, David Raunig, Constantine Gatsonis, Erich Huang, Marina Kondratovich, Lisa Mcshane, Anthony Reeves, Daniel Barboriak, Alexander Guimaraes, and Richard Wahl. Metrology standards for quantitative imaging biomarkers. *Radiology*, 277:142202, 08 2015.

[11] Robert Gillies, Paul Kinahan, and Hedvig Hricak. Radiomics: Images are more than pictures, they are data. *Radiology*, 278:151169, 11 2015.

[12] Meghan Lubner, Andrew Smith, Kumar Sandrasegaran, Dushyant Sahani, and Perry Pickhardt. Ct texture analysis: Definitions, applications, biologic correlates, and challenges. *Radiographics*, 37:1483–1503, 09 2017.

[13] Burak Kocak, Emine Durmaz, Cagri Erdim, Ece Ates Kus, Ozlem Kaya, and Ozgur Kilickesmez. Radiomics of renal masses: Systematic review of reproducibility and validation strategies. *American Journal of Roentgenology*, pages 1–8, 10 2019.

[14] Burak Kocak, Emine Sebnem Durmaz, Ece Ates, and Melis Baykara Ulusan. Radiogenomics in clear cell renal cell carcinoma: Machine learning–based high-dimensional quantitative ct texture analysis in predicting pbrm1 mutation status. *American Journal of Roentgenology*, 212(3):W55–W63, 2019. PMID: 30601030.

[15] Rodrigo Suarez-Ibarrola, Simon Hein, Gerd Reis, Christian Gratzke, and Arkadiusz Miernik. Current and future applications of machine and deep learning in urology: a review of the literature on urolithiasis, renal cell carcinoma, and bladder and prostate cancer. *World Journal of Urology*, 38, 10 2020.

[16] Luca Saba, Mainak Biswas, Venkatanareshbabu Kuppili, Elisa Cuadrado Godia, Harman S. Suri, Damodar Reddy Edla, Tomaž Omerzu, John R. Laird, Narendra N. Khanna, Sophie Mavrogeni, Athanasios Protogerou, Petros P. Sfikakis, Vijay Viswanathan, George D. Kitas, Andrew Nicolaides, Ajay Gupta, and Jasjit S. Suri. The present and future of deep learning in radiology. *European Journal of Radiology*, 114:14–24, 2019.

[17] Shijun Wang and Ronald M. Summers. Machine learning and radiology. *Medical Image Analysis*, 16(5):933–951, 2012.

[18] Yann LeCun, Y. Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521:436–44, 05 2015.

[19] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen A W M van der Laak, Bram van Ginneken, and Clara I Sánchez. A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42:60–88, 2017.

[20] Sheng-Chun Huang, Avinash Pareek, Shreya Seyyedi, Ishan Banerjee, and Matthew P. Lungren. Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines. *npj Digital Medicine*, 3:136, 2020.

[21] Sören Richard Stahlschmidt, Benjamin Ulfenborg, and Jane Synnergren. Multimodal deep learning for biomedical data fusion: a review. *Briefings in bioinformatics*, 23(2):bbab569, 2022.

[22] Dhruv Sharma, Sanjay Purushotham, and Chandan K. Reddy. MedFuseNet: An attention-based multimodal deep learning model for visual question answering in the medical domain. *Scientific Reports*, 11(1):19826, 2021.

[23] Jing Wu, Yuli Wang, Zhusi Zhong, Weihua Liao, Natalia Trayanova, Zhicheng Jiao, and Harrison X. Bai. Vision-language foundation model for 3d medical imaging. *npj Artificial Intelligence*, 1:17, 2025.

[24] Tugba Akinci D'Antonoli, Christian Bluethgen, Renato Cuocolo, Michail E. Klontzas, Andrea Ponsiglione, and Burak Kocak. Foundation models for radiology: fundamentals, applications, opportunities, challenges, risks, and prospects. *Diagnostic and Interventional Radiology*, 2025. Epub 8 July 2025, DOI: 10.4274/dir.2025.253445.

[25] Mehmet Yigit Avci, Pedro Borges, Paul Wright, Mehmet Yigitsoy, Sebastien Ourselin, and Jorge Cardoso. MR-CLIP: Efficient metadata-guided learning of MRI contrast representations. *arXiv preprint arXiv:2507.00043*, 2025.

[26] Sinyoung Ra, Jonghun Kim, Inye Na, Eun Sook Ko, and Hyunjin Park. Enhancing radiomics features via a large language model for classifying benign and malignant breast tumors in mammography. *Computer Methods and Programs in Biomedicine*, 265:108765, 2025.

[27] R. Khan, C. Chen, A. Zaman, J. Wu, H. Mai, L. Su, Y. Kang, B. Huang, et al. RenalSegNet: automated segmentation of renal tumor, veins, and arteries in contrast-enhanced CT scans. *Complex & Intelligent Systems*, 11:131, 2025.

[28] Seokhwan Bang, Heehwan Wang, Hoyoung Bae, Sung-Hoo Hong, Jiook Cha, and Moon Hyung Choi. Computed Tomography-Based Radiomics Diagnostic Model for Fat-Poor Small Renal Tumor Subtypes. *Diagnostics*, 15(11):1365, 2025.

[29] Annemarie Uhlig, Johannes Uhlig, Andreas Leha, Lorenz Biggemann, Sophie Bachanek, Michael Stöckle, Mathias Reichert, Joachim Lotz, Philip Zeuschner, and Alexander Maßmann. Radiomics and machine learning for renal tumor subtype assessment using multiphase computed tomography in a multicenter setting. *European Radiology*, 34:6254–6263, 2024.

[30] Ali Sheikhy, Fatemeh Dehghani Firouzabadi, Nathan Lay, Negin Jarrah, Pouria Yazdian Anari, and Ashkan Malayeri. State of the art review of AI in renal imaging. *Abdominal Radiology*, 2025.

[31] Nicholas Heller, Fabian Isensee, Klaus Maier-Hein, Xiaoshuai Hou, Chunmei Xie, Fengyi Li, Yang Nan, Guangrui Mu, Zhiyong Lin, Miofei Han, Guang Yao, Yaozong Gao, Yao Zhang, Yixin Wang, Feng Hou, Jiawei Yang, Guangwei Xiong, Jiang Tian, Cheng Zhong, and Christopher Weight. The state of the art in kidney and kidney tumor segmentation in contrast-enhanced ct imaging: Results of the kits19 challenge. *Medical Image Analysis*, 67, 10 2020.

[32] John N Weinstein, Eric A Collisson, Gordon B Mills, Kenna R Shaw, Brad A Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, and Joshua M Stuart. The cancer genome atlas pan-cancer analysis project. *Nature genetics*, 45(10):1113–1120, 2013.

[33] Kenneth Clark, Bruce Vendt, Kirk Smith, John Freymann, Justin Kirby, Paul Koppel, Stephen Moore, Stanley Phillips, David Maffitt, Michael Pringle, et al. The cancer imaging archive (tcia): maintaining and operating a public information repository. *Journal of digital imaging*, 26:1045–1057, 2013.

[34] Nicholas Heller, Niranjan Sathianathen, Arveen Kalapara, Edward Walczak, Keenan Moore, Heather Kaluzniak, Joël Rosenberg, Paul Blake, Zachary Rengel, Makinna Oestreich, Joshua Dean, Michael Tradewell, Aneri Shah, Resha Tejpaul, Zachary Edgerton, Matthew Peterson, Shaneabbas Raza, Subodh Regmi, Nikolaos Papanikolopoulos, and Christopher Weight. The kits19 challenge data: 300 kidney tumor cases with clinical context, ct semantic segmentations, and surgical outcomes, 03 2019.

[35] Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.

[36] Daniel Wolf, Tristan Payer, Catharina Silvia Lisson, Christoph Gerhard Lisson, Meinrad Beer, Michael Götz, and Timo Ropinski. Self-supervised pre-training with contrastive and masked autoencoder methods for dealing with small datasets in deep learning for medical imaging. *Scientific Reports*, 13(1):20260, 2023.

[37] Nima Tajbakhsh, Jae Y Shin, Suryakanth R Gurudu, R Todd Hurst, Christopher B Kendall, Michael B Gotway, and Jianming Liang. Convolutional neural networks for medical image analysis: Full training or fine tuning? *IEEE transactions on medical imaging*, 35(5):1299–1312, 2016.

[38] Hee E Kim, Alejandro Cosa-Linan, Nandhini Santhanam, Mahboubeh Jannesari, Mate E Maros, and Thomas Ganslandt. Transfer learning for medical image classification: a literature review. *BMC medical imaging*, 22(1):69, 2022.

[39] Shekoofeh Azizi, Laura Culp, Jan Freyberg, Basil Mustafa, Sebastien Baur, Simon Kornblith, Ting Chen, Nenad Tomasev, Jovana Mitrović, Patricia Strachan, et al. Robust and data-efficient generalization of self-supervised machine learning for diagnostic imaging. *Nature Biomedical Engineering*, 7(6):756–779, 2023.

[40] Gulay Maçin, Fatih Genç, Burak Taşcı, Sengul Dogan, and Turker Tuncer. Kidneynext: a lightweight convolutional neural network for multi-class renal tumor classification in computed tomography imaging. *Journal of Clinical Medicine*, 14(14):4929, 2025.

[41] Jeff Van Oss, Gowtham Krishnan Murugesan, Diana McCrumb, and Rahul Soni. Image segmentations produced by bamf under the aimi annotations initiative. *(No Title)*, 2023.

[42] Joost J.M. van Griethuysen, Andriy Fedorov, Chintan Parmar, Ahmed Hosny, Nicole Aucoin, Vivek Narayan, Regina G.H. Beets-Tan, Jean-Christophe Fillion-Robin, Steve Pieper, and Hugo J.W.L. Aerts. Computational radiomics system to decode the radiographic phenotype. *Cancer Research*, 77(21):e104–e107, 10 2017.

[43] Jundong Li, Kewei Cheng, Suhang Wang, Fred Morstatter, Robert P. Trevino, Jiliang Tang, and Huan Liu. Feature selection: A data perspective. *ACM Comput. Surv.*, 50(6), December 2017.

[44] Arif Priyatno and Triyanna Widiyaningtyas. A systematic literature review: Recursive feature elimination algorithms. *JITK (Jurnal Ilmu Pengetahuan dan Teknologi Komputer)*, 9:196–207, 02 2024.

[45] Yueyan Bian, Jin Li, Chuyang Ye, Xiuqin Jia, and Qi Yang. Artificial intelligence in medical imaging: From task-specific models to large-scale foundation models. *Chinese Medical Journal*, 138(06):651–663, 2025.

[46] Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. Segment anything in medical images. *Nature Communications*, 15(1):654, 2024.

[47] Hao Guan and Mingxia Liu. Domain adaptation for medical image analysis: a survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2021.

[48] Dow-Mu Koh, Nickolas Papanikolaou, Ulrich Bick, Rowland Illing, Charles E Kahn Jr, Jayshree Kalpathi-Cramer, Celso Matos, Luis Martí-Bonmatí, Anne Miles, Seong Ki Mun, et al. Artificial intelligence and machine learning in cancer imaging. *Communications Medicine*, 2(1):133, 2022.

[49] Cheng-Jiang Wei, Yan Tang, Yang-Bai Sun, Tie-Long Yang, Cheng Yan, Hui Liu, Jun Liu, Jing-Ning Huang, Ming-Han Wang, Zhen-Wei Yao, et al. A multicenter study of neurofibromatosis type 1 utilizing deep learning for whole body tumor identification. *npj Digital Medicine*, 8(1):56, 2025.

[50] Saqib Ali, Jianqiang Li, Yan Pei, Rooha Khurram, Khalil ur Rehman, and Abdul Basit Rasool. State-of-the-art challenges and perspectives in multi-organ cancer diagnosis via deep learning-based methods. *Cancers*, 13(21):5546, 2021.

[51] Shaoting Zhang and Dimitris Metaxas. On the challenges and perspectives of foundation models for medical image analysis. *Medical image analysis*, 91:102996, 2024.

[52] Farida Mohsen, Hazrat Ali, Nady El Hajj, and Zubair Shah. Artificial intelligence-based methods for fusion of electronic health records and imaging data. *Scientific Reports*, 12(1):17981, 2022.

[53] Zhengyang Sun et al. A scoping review on multimodal deep learning in radiology. *Journal of Biomedical Informatics*, 2023.

[54] Imran Ul Haq et al. Advancements in medical radiology through multimodal machine learning: A comprehensive overview. *Bioengineering*, 12(5):477, 2025.

[55] Amara Tariq, Imon Banerjee, Hari Trivedi, and Judy Gichoya. Multimodal artificial intelligence models for radiology. *BJR|Artificial Intelligence*, 2(1):ubae017, 01 2025.

[56] Yihao Li, Mostafa El Habib Daho, Pierre-Henri Conze, Rachid Zeghlache, Hugo Le Boité, Ramin Tadayoni, Béatrice Cochener, Mathieu Lamard, and Gwenolé Quellec. A review of deep learning-based information fusion techniques for multimodal medical image classification. *Computers in Biology and Medicine*, 177:108635, 2024.

[57] Benjamin D. Simon, Koray B. Ozyoruk, David G. Gelikman, S. A. Harmon, and B. Türkbey. The future of multimodal artificial intelligence models for integrating imaging and clinical metadata: A narrative review. *Diagnostic and Interventional Radiology*, 2025.

[58] Adrienne Kline, Hanyin Wang, Yikuan Li, Saya Dennis, Meghan Hutch, Zhenxing Xu, Fei Wang, Feixiong Cheng, and Yuan Luo. Multimodal machine learning in precision health: A scoping review. *NPJ digital medicine*, 5(1):171, 2022.

[59] Pegah Kharazmi, Sunil Kalia, Harvey Lui, ZJ Wang, and TK Lee. A feature fusion system for basal cell carcinoma detection through data-driven feature learning and patient profile. *Skin research and technology*, 24(2):256–264, 2018.

[60] Jordan Yap, William Yolland, and Philipp Tschandl. Multimodal skin lesion classification using deep learning. *Experimental dermatology*, 27(11):1261–1267, 2018.

[61] Adam Yala, Constance Lehman, Tal Schuster, Tally Portnoi, and Regina Barzilay. A deep learning mammography-based model for improved breast cancer risk prediction. *Radiology*, 292(1):60–66, 2019.

[62] Qian Sun, Lei Yu, Zhongquan Song, Can Wang, Wei Li, Wang Chen, Juan Xu, and Shuhua Han. Deep learning and radiomics fusion for predicting the invasiveness of lung adenocarcinoma within ground glass nodules. *Scientific Reports*, 15(1):29285, 2025.

[63] Yi Xiao, Jun Zhang, Cheng Chi, and Chunyu Wang. A novel ed triage framework using conditional imputation, multi-scale semantic learning, and cross-modal fusion. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 13–22. Springer, 2025.

[64] Min-Yen Hsu, Jeng-Yuan Chiou, Jung-Tzu Liu, Chee-Ming Lee, Ya-Wen Lee, Chien-Chih Chou, Shih-Chang Lo, Edy Kornelius, Yi-Sun Yang, Sung-Yen Chang, et al. Deep learning for automated diabetic retinopathy screening fused with heterogeneous data from ehrs can lead to earlier referral decisions. *Translational Vision Science & Technology*, 10(9):18–18, 2021.

[65] Lili He, Hailong Li, Ming Chen, Jinghua Wang, Mekibib Altaye, Jonathan R Dillman, and Nehal A Parikh. Deep multimodal learning from mri and clinical data for early prediction of neurodevelopmental deficits in very preterm infants. *Frontiers in neuroscience*, 15:753033, 2021.

[66] Declan Grant, Bartłomiej W Papież, Guy Parsons, Lionel Tarassenko, and Adam Mahdi. Deep learning classification of cardiomegaly using combined imaging and non-imaging icu data. In *Annual Conference on Medical Image Understanding and Analysis*, pages 547–558. Springer, 2021.

[67] Zeynel A Samak, Philip Clatworthy, and Majid Mirmehdi. Prediction of thrombectomy functional outcomes using multimodal data. In *Annual Conference on Medical Image Understanding and Analysis*, pages 267–279. Springer, 2020.

[68] Ziyi Liu, Jiaqi Zhang, Yongshuai Hou, Xinran Zhang, Ge Li, and Yang Xiang. Machine learning for multimodal electronic health records-based research: Challenges and perspectives. In *China Health Information Processing Conference*, pages 135–155. Springer, 2022.

[69] Jeremy Kawahara, Sara Daneshvar, Giuseppe Argenziano, and Ghassan Hamarneh. Seven-point checklist and skin lesion classification using multitask multimodal neural nets. *IEEE journal of biomedical and health informatics*, 23(2):538–546, 2018.

[70] Firas Khader, Gustav Müller-Franzes, Tianci Wang, Tianyu Han, Soroosh Tayebi Arasteh, Christoph Haarburger, Johannes Stegmaier, Keno Bressem, Christiane Kuhl, Sven Nebelung, et al. Multimodal deep learning for integrating chest radiographs and clinical parameters: a case for transformers. *Radiology*, 309(1):e230806, 2023.

[71] Hong-Yu Zhou, Yizhou Yu, Chengdi Wang, Shu Zhang, Yuanxu Gao, Jia Pan, Jun Shao, Guangming Lu, Kang Zhang, and Weimin Li. A transformer-based representation-learning model with unified processing of multimodal input for clinical diagnostics. *Nature biomedical engineering*, 7(6):743–755, 2023.

[72] Firas Khader, Jakob Nikolas Kather, Gustav Müller-Franzes, Tianci Wang, Tianyu Han, Soroosh Tayebi Arasteh, Karim Hamesch, Keno Bressem, Christoph Haarburger, Johannes Stegmaier, et al. Medical transformer for multimodal survival prediction in intensive care: integration of imaging and non-imaging data. *Scientific Reports*, 13(1):10666, 2023.

[73] Shangran Qiu, Gary H Chang, Marcello Panagia, Deepa M Gopal, Rhoda Au, and Vijaya B Kolachalama. Fusion of deep learning models of mri scans, mini–mental state examination, and logical memory test enhances diagnosis of mild cognitive impairment. *Alzheimer's & Dementia: Diagnosis, Assessment & Disease Monitoring*, 10:737–749, 2018.

[74] Shih-Cheng Huang, Anuj Pareek, Roham Zamanian, Imon Banerjee, and Matthew P Lungren. Multimodal fusion with deep neural networks for leveraging ct imaging and electronic health record: a case-study in pulmonary embolism detection. *Scientific reports*, 10(1):22147, 2020.

[75] Guoxiu Lu, Ronghui Tian, Wei Yang, Ruibo Liu, Dongmei Liu, Zijie Xiang, and Guoxu Zhang. Deep learning radiomics based on multimodal imaging for distinguishing benign and malignant breast tumours. *Frontiers in Medicine*, 11:1402967, 2024.

[76] Jiseong Ryu et al. Vision–language foundation models for medical imaging. *Biomedicines & Pharmacotherapy (Springer) / Biomedical Engineering Letters (Springer) – article page*, 2025. Review of medical VLMs 2022–2024.

[77] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.

[78] Benedikt Boecking, Naoto Usuyama, Shruthi Bannur, Daniel C. Castro, Anton Schwaighofer, Stephanie Hyland, Maria Wetscherek, Tristan Naumann, Aditya Nori, Javier Alvarez-Valle, Hoifung Poon, and Ozan Oktay. Making the most of text semantics to improve biomedical vision–language processing. *arXiv preprint arXiv:2204.09817*, 2022.

[79] Daoming Lyu, Xingbo Wang, Yong Chen, and Fei Wang. Language model and its interpretability in biomedicine: A scoping review. *Iscience*, 27(4), 2024.

[80] Shih-Cheng Huang, Liyue Shen, Matthew P Lungren, and Serena Yeung. Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3942–3951, 2021.

[81] Taesung Park, Alexei A Efros, Richard Zhang, and Jun-Yan Zhu. Contrastive learning for unpaired image-to-image translation. In *European conference on computer vision*, pages 319–345. Springer, 2020.

[82] Zifeng Wang, Zhenbang Wu, Dinesh Agarwal, and Jimeng Sun. Medclip: Contrastive learning from unpaired medical images and text. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing*, volume 2022, page 3876, 2022.

[83] Songyang Zhang et al. Large-scale domain-specific pretraining for biomedical vision–language processing. *arXiv preprint arXiv:2303.00915*, 2023.

[84] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, and Ronald M Summers. Tienet: Text-image embedding network for common thorax disease classification and reporting in chest x-rays. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 9049–9058, 2018.

[85] Ian Hartsock et al. Vision–language models for medical report generation and visual question answering: A review. *Frontiers in Artificial Intelligence*, 2024.

[86] Omkar Thawkar, Abdelrahman Shaker, Sahal Shaji Mullappilly, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Jorma Laaksonen, and Fahad Shahbaz Khan. Xray-gpt: Chest radiographs summarization using medical vision-language models. *arXiv preprint arXiv:2306.07971*, 2023.

[87] Sadaf Khademi, Mehran Shabanpour, Reza Taleei, Anastasia Oikonomou, and Arash Mohammadi. Autorad-lung: A radiomic-guided prompting autoregressive vision-language model for lung nodule malignancy prediction. *arXiv preprint arXiv:2503.20662*, 2025.

[88] Yanyan Huang, Wanli Zhang, Peixiang Huang, Yu Fu, Ruimeng Yang, and Lequan Yu. Bridging radiological images and factors with vision–language model for accurate diagnosis of proliferative hepatocellular carcinoma. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2025*, 2025. Open access version.

[89] Chenlin Du, Xiaoxuan Chen, Jingyi Wang, Junjie Wang, Zhongsen Li, Zongjiu Zhang, and Qicheng Lao. Prompting vision-language models for dental notation aware abnormality detection. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 687–697. Springer, 2024.

[90] Ming Y. Lu, Bowen Chen, Andrew Zhang, Drew F. K. Williamson, Richard J. Chen, Tong Ding, Long Phi Le, Yung-Sung Chuang, and Faisal Mahmood. Visual language pretrained multiple instance zero-shot transfer for histopathology images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19764–19775, 2023.

[91] Jiangbo Shi, Chen Li, Tieliang Gong, Yefeng Zheng, and Huazhu Fu. Vila-mil: Dual-scale vision-language multiple instance learning for whole slide image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11248–11258, 2024.

[92] Jiazhen Pan, Che Liu, Junde Wu, Fenglin Liu, Jiayuan Zhu, Hongwei Bran Li, Chen Chen, Cheng Ouyang, and Daniel Rueckert. Medvlm-r1: Incentivizing medical reasoning capability of vision-language models (vlms) via reinforcement learning. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 337–347. Springer, 2025.

[93] Manav Nitin Kapadnis, Sohan Patnaik, Abhilash Nandy, Sourjyadip Ray, Pawan Goyal, and Debdoot Sheet. Serpent-vlm: Self-refining radiology report generation using vision language models. *arXiv preprint arXiv:2404.17912*, 2024.

[94] Yoojin Nam, Dong Yeong Kim, Sunggu Kyung, Jinyoung Seo, Jeong Min Song, Jimin Kwon, Jihyun Kim, Wooyoung Jo, Hyungbin Park, Jimin Sung, et al. Multimodal large language models in medical imaging: Current state and future directions. *Korean Journal of Radiology*, 26(10):900, 2025.

[95] Juan M. Zambrano Chaves et al. A clinically accessible small multimodal radiology model. *Nature Communications*, 2025.

[96] Yijing Zhou, Yadong Lu, Qingli Li, Xinxing Li, and Yan Wang. Deep association multimodal learning for zero-shot spatial transcriptomics prediction. pages 131–140, 09 2025.

[97] Rhydian Windsor, Amir Jamaludin, Timor Kadir, and Andrew Zisserman. Vision-language modelling for radiological imaging and reports in the low data regime. *arXiv preprint arXiv:2303.17644*, 2023.

[98] Felix Krones, Umar Marikkar, Guy Parsons, Adam Szmul, and Adam Mahdi. Review of multimodal machine learning approaches in healthcare. *Information Fusion*, 114:102690, 2025.

[99] Burcu Oltu, Selda Güney, Seniha Esen Yuksel, and Berna Dengiz. Automated classification of chest x-rays: a deep learning approach with attention mechanisms. *BMC Medical Imaging*, 25(1):71, 2025.

[100] Md Rabiul Hasan, Shah Muhammad Azmat Ullah, and Sheikh Md Rabiul Islam. Recent advancement of deep learning techniques for pneumonia prediction from chest x-ray image, 2024.

[101] Xiao Fang, Yi Lin, Dong Zhang, Kwang-Ting Cheng, and Hao Chen. Aligning medical images with general knowledge from large language models. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 57–67. Springer, 2024.

[102] Guang Yang, Qinghao Ye, and Jun Xia. Unbox the black-box for the medical explainable ai via multi-modal and multi-centre data fusion: A mini-review, two showcases and beyond. *Information Fusion*, 77:29–52, 2022.

[103] Yueqin Diao et al. Prior-guided prototype aggregation learning for alzheimer's disease diagnosis. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2025*, 2025.

[104] Rami Vanguri, Jia Luo, Andrew Aukerman, Jacklynn Egger, Christopher Fong, Natally Horvat, Andrew Pagano, Jose Araujo-Filho, Luke Geneslaw, Hira Rizvi, Ramon Sosa, Kevin Boehm, Soo-Ryum Yang, Francis Bodd, Katia Ventura, Travis Hollmann, Michelle Ginsberg, Jianjiong Gao, Rami Vanguri, and Sohrab Shah. Multimodal integration of radiology, pathology and genomics for prediction of response to pd-(l)1 blockade in patients with non-small cell lung cancer. *Nature Cancer*, 3:1–14, 08 2022.

[105] Gwénolé Abgrall, Andre L Holder, Zaineb Chelly Dagdia, Karine Zeitouni, and Xavier Monnet. Should ai models be explainable to clinicians? *Critical Care*, 28(1):301, 2024.

[106] Haomin Chen, Catalina Gomez, Chien-Ming Huang, and Mathias Unberath. Explainable medical imaging ai needs human-centered design: guidelines and evidence from a systematic review. *NPJ digital medicine*, 5(1):156, 2022.

[107] Essam H Houssein, Amr M Gamal, Eman MG Younis, and Ebtsam Mohamed. Explainable artificial intelligence for medical imaging systems using deep learning: a comprehensive review. *Cluster Computing*, 28(7):469, 2025.

[108] Yashbir Singh, Quincy A Hathaway, Varekan Keishing, Sara Salehi, Yujia Wei, Natally Horvat, Diana V Vera-Garcia, Ashok Choudhary, Almurtadha Mula Kh, Emilio Quaia, et al. Beyond post hoc explanations: A comprehensive framework for accountable ai in medical imaging through transparency, interpretability, and explainability. *Bioengineering*, 12(8):879, 2025.

[109] Mateja Napravnik, Franko Hržić, Martin Urschler, Damir Miletić, and Ivan Štajduhar. Lessons learned from radiologynet foundation models for transfer learning in medical radiology. *Scientific Reports*, 15(1):21622, 2025.

[110] M Sree Vani, Rayapati Venkata Sudhakar, A Mahendar, Sukanya Ledalla, Marepalli Radha, and M Sunitha. Personalized health monitoring using explainable ai: bridging trust in predictive healthcare. *Scientific Reports*, 15(1):31892, 2025.

[111] William Thomas Hrinivich, Tonghe Wang, and Chunhao Wang. Interpretable and explainable machine learning models in oncology, 2023.

[112] Sahil Sharma, Muskaan Singh, Liam McDaid, and Saugat Bhattacharyya. Xai-based data visualization in multimodal medical data. *bioRxiv*, pages 2025–07, 2025.

[113] Friederike Jungmann, Sebastian Ziegelmayer, Fabian K Lohoefer, Stephan Metz, Christina Müller-Leisse, Maximilian Englmaier, Marcus R Makowski, Georgios A Kaissis, and Rickmer F Braren. Algorithmic transparency and interpretability measures improve radiologists' performance in bi-rads 4 classification. *European Radiology*, 33(3):1844–1851, 2023.

[114] Aurélie Pahud de Mortanges, Haozhe Luo, Shelley Zixin Shu, Amith Kamath, Yannick Suter, Mohamed Shelan, Alexander Pöllinger, and Mauricio Reyes. Orchestrating explainable artificial intelligence for multimodal and longitudinal data in medical imaging. *NPJ digital medicine*, 7(1):195, 2024.

[115] Weina Jin, Xiaoxiao Li, Mostafa Fatehi, and Ghassan Hamarneh. Guidelines and evaluation of clinical explainable ai in medical image analysis. *Medical image analysis*, 84:102684, 2023.

[116] Xavier P. Burgos-Artizzu, David Coronado-Gutiérrez, Brenda Valenzuela-Alcaraz, Elisenda Bonet-Carne, Elisenda Eixarch, Fatima Crispi, and Eduard Gratacós. Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes. *Scientific Reports*, 10(1):10200, December 2020.

[117] Thomas L. A. van den Heuvel, Dagmar de Bruijn, Chris L. de Korte, and Bram van Ginneken. Automated measurement of fetal head circumference using 2d ultrasound images. *PLOS ONE*, 13(8):1–20, August 2018.

[118] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11966–11976, 2022.

[119] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015.

[120] Ze Liu, Han Hu, Yutong Lin, Zhuliang Yao, Zhenda Xie, Yixuan Wei, Jia Ning, Yue Cao, Zheng Zhang, Li Dong, Furu Wei, and Baining Guo. Swin transformer v2: Scaling up capacity and resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12009–12019, June 2022.

[121] Mingxing Tan and Quoc Le. Efficientnetv2: Smaller models and faster training. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 10096–10106. PMLR, 18–24 Jul 2021.

[122] Andrew Howard, Mark Sandler, Bo Chen, Weijun Wang, Liang-Chieh Chen, Mingxing Tan, Grace Chu, Vijay Vasudevan, Yukun Zhu, Ruoming Pang, Hartwig Adam, and Quoc Le. Searching for mobilenetv3. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1314–1324, 2019.

[123] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*, 2021.

[124] Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *European Conference on Computer Vision (ECCV)*, pages 459–479. Springer, 2022.

[125] M Jorge Cardoso, Wenqi Li, Richard Brown, Nic Ma, Eric Kerfoot, Yiheng Wang, Benjamin Murrey, Andriy Myronenko, Can Zhao, Dong Yang, et al. Monai: An open-source framework for deep learning in healthcare. *arXiv preprint arXiv:2211.02701*, 2022.

[126] Xuerui Qiu, Rui-Jie Zhu, Yuhong Chou, Zhaorui Wang, Liang-jian Deng, and Guoqi Li. Gated attention coding for training high-performance and efficient spiking neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 601–610, 2024.

[127] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7132–7141, 2018.

[128] Dennis Sonnier, W. Sawyer, John Seal, Colin Curtis, Jack McGee, Alec Slayden, and Korak Sarkar. Three-dimensional visualization with virtual reality facilitates complex live donor renal transplant. *Ochsner Journal*, 22, 07 2022.